AUTONOMOUS UNIVERSITY OF CIUDAD JUÁREZ

MULTIDISCIPLINARY DIVISION OF CIUDAD UNIVERSITARIA

DEPARTMENT OF INDUSTRIAL AND MANUFACTURING ENGINEERING

Pen Plotter CNC

Motors and Controllers

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1 Introduction

In this project, we will build a simple CNC machine to draw the logo of the Autonomous University of Ciudad Juárez (UACJ) on paper. We will use stepper motors to move a pencil, pen, or marker with precision, controlled by an Arduino board and configured with the respective Marlin firmware.

2 Objectives

Design and build a CNC machine controlled by Arduino and Marlin firmware, capable of autonomously drawing the UACJ logo on paper, using two stepper motors for the x and y axes, and a servomotor for the z axis.

2.1 Specific Objectives

- Design the mechanical structure of the CNC, ensuring stability and precision.
- Implement stepper motor control using Arduino.
- Configure the system using Marlin firmware.
- Generate G-code from a design in Inkscape.
- Test and calibrate the machine to ensure proper operation and precision.

3 Theoretical Framework

3.1 What is a CNC machine?

CNC (Computer Numerical Control) is an automated machine system that works with programmed commands or functions, whose instructions are stored on a storage disk. It is a machine tool used for:

- Machining parts from various materials.
- Automated processes.
- Computer programming.
- Computer control.

4 Requirements

4.1 Hardware

4.1.1 Microcontroller

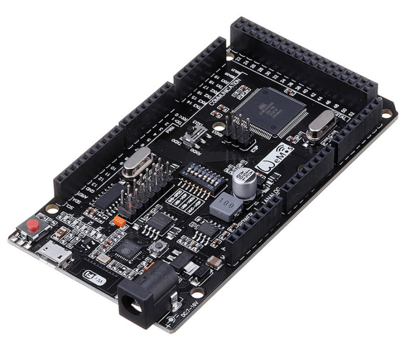

Figure 1: WeMos Mega WiFi R3 ATmega2560 + ESP8266 32MB.

The Arduino is an open-source electronic board based on a reprogrammable microcontroller. It is designed to facilitate the development of electronic projects through the connection of various sensors, actuators, and other devices through its digital and analog input and output pins.

4.1.2 Expansion Board

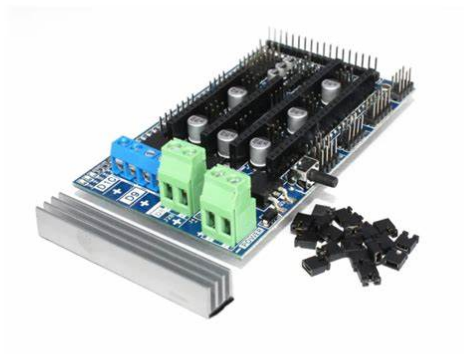

Figure 2: Ramps 1.6.

The **Ramps 1.6** expansion board is designed to be mounted on an **Arduino Mega 2560**. Its main function is to facilitate the connection and control of up to five stepper motors independently, corresponding to the X, Y, Z axes and two extruders (E0 and E1). This board is commonly used in 3D printers and CNC machines, as it allows the integration of drivers such as **A4988** or **DRV8825**, in addition to other components such as switches, fans, and thermistors.

4.1.3 Stepper Motor

Figure 3: 17HD48002H-22B.

It is a **NEMA 17** type stepper motor, ideal for precision applications such as CNC machines. It offers a step angle of 1.8° (200 steps per revolution) and operates with a current of 1.5 A per phase. Its compact size and good torque make it suitable for moving the x, y, and z axes of the project with high precision.

4.1.4 Servomotor

Figure 4: SG90.

The SG90 servomotor is a compact and lightweight actuator commonly used in robotics. It operates on 4.8–6V and rotates about 180°, controlled by PWM signals. In this project, it lifts and lowers the pen, allowing the CNC to draw or move without marking the paper.

4.1.5 Display Screen

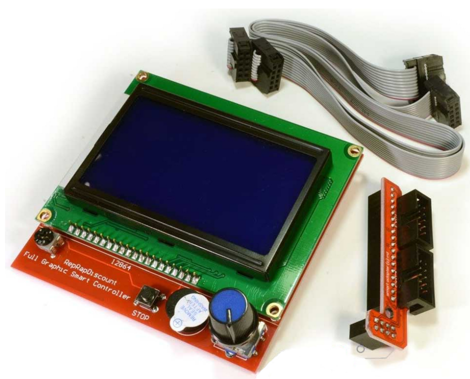

Figure 5: RepRapDiscount 12864.

It is a complete module that includes a 128×64 pixel dot matrix LCD screen, an SD card reader, and a rotary encoder. This device allows controlling a 3D printer or CNC machine autonomously, without the need for a computer, displaying process information and facilitating navigation through the firmware menu.

4.1.6 Drivers

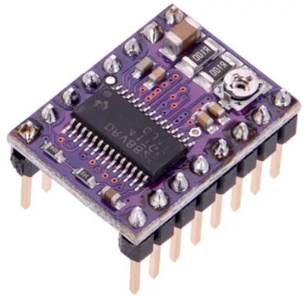

Figure 6: DRV8825.

The DRV8825 driver is a bipolar stepper motor controller, which is used to control stepper motors, especially in applications that require high precision and movement resolution, such as 3D printers and CNC machines.

4.1.7 Adjustable Power Supply

Figure 7: Korad KD3005D.

The **Korad KD3005D** power supply is a versatile instrument designed for various electronic applications. It features an adjustable voltage range of up to 30V and a maximum current of 5A.

4.2 Software

4.2.1 Marlin Firmware

Figure 8: Marlin Firmware.

Marlin is an open-source firmware widely used to control 3D printers, CNC machines, laser engravers, and plotters. It is responsible for interpreting G-code commands and converting them into physical movements, controlling motors, servos, and sensors. It is compatible with boards such as RAMPS or SKR, and allows easy configuration of parameters such as axes, limits, and servomotors.

4.2.2 PlatformIO

Figure 9: PlatformIO.

PlatformIO is an open-source development environment used with Visual Studio Code to program microcontrollers. It allows compiling and uploading firmware like Marlin quickly and in an organized manner.

4.2.3 VSCode

Figure 10: VSCode.

Visual Studio Code (VS Code) is a lightweight and cross-platform code editor developed by Microsoft. It is widely used by programmers thanks to its speed, customization, and support for extensions.

4.2.4 Inkscape

Figure 11: Inkscape.

Inkscape is an open-source vector graphics design software, used to create and edit images in SVG format.

4.2.5 Pronterface

Figure 12: Pronterface.

Pronterface is a graphical interface of the Printrun program, used to control 3D printers and CNC machines from a computer. It allows sending G-code commands, moving axes, controlling temperature (in 3D printers), and monitoring the machine's status in real time.

5 Implementation

5.1 Structure

We had an already built structure made of pressed wood, which was adapted according to the needs that arose during the project development. In particular, modifications were made to the axes and the arrangement of the motors within the structure was changed.

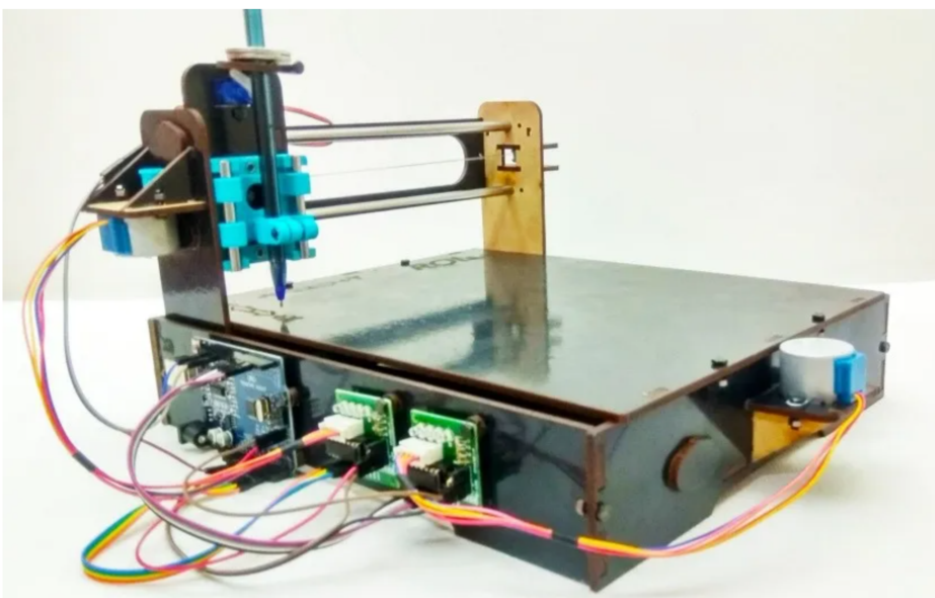

Figure 13: Mini CNC Plotter Kit.

5.2 Motor Assembly

The structure design was created in **SolidWorks** according to the requested structure.

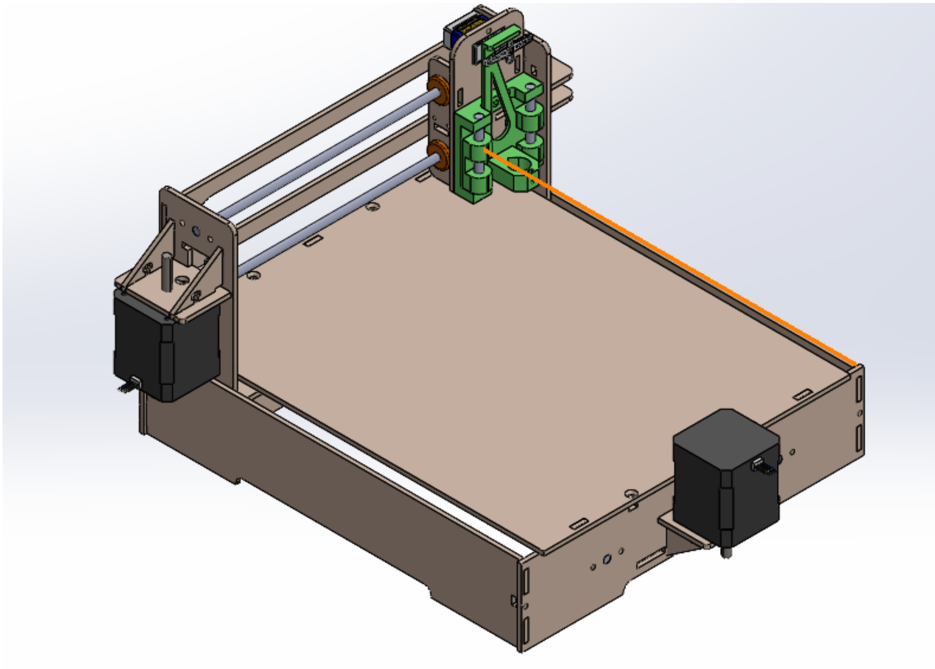

Figure 14: SolidWorks Design.

5.3 Pictorial Diagram

A diagram was created that represents the project connections, visually showing how the used components are interconnected.

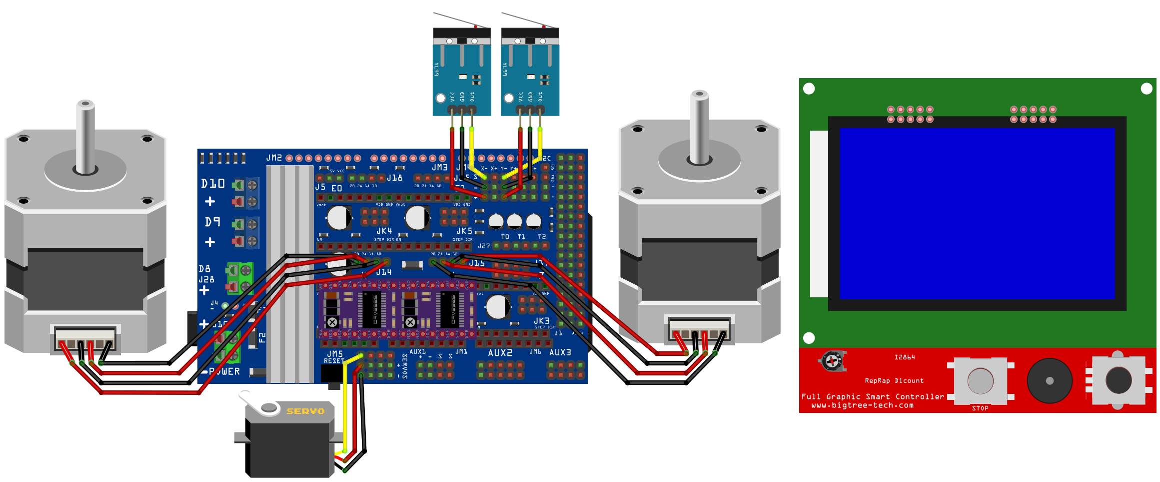

Figure 15: Diagram made in Fritzing.

5.4 Motor Testing and Connection

After generating the G-code, a test was performed with the **NEMA 17** stepper motors. Using the **RepRapDiscount** controller, it was possible to start and stop work directly from the SD card. Additionally, this controller allows easy adjustment of parameters such as motor position and speed.

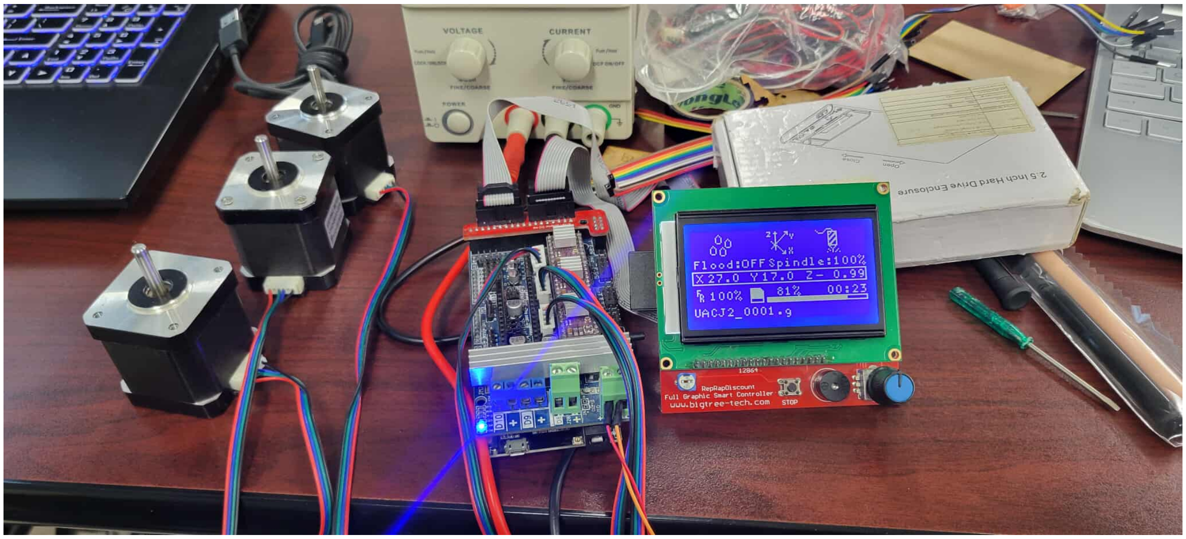

Figure 16: Motor connections with Ramps 1.6.

5.5 G-code Generation

During the project, **Inkscape** software was used to design vector graphics. Then, with the help of the **J Tech Photonics Laser Tool** extension, we converted these designs into G-code, which can be interpreted by the intelligent graphic controller **RepRapDiscount** 12864. This controller works together with stepper motors and **Marlin** firmware, allowing the motors to perform the necessary movements for engraving with precision.

At the bottom, the *offset* is defined, which corresponds to the distance difference between the limit switch and the tool tip (in this case, a pen).

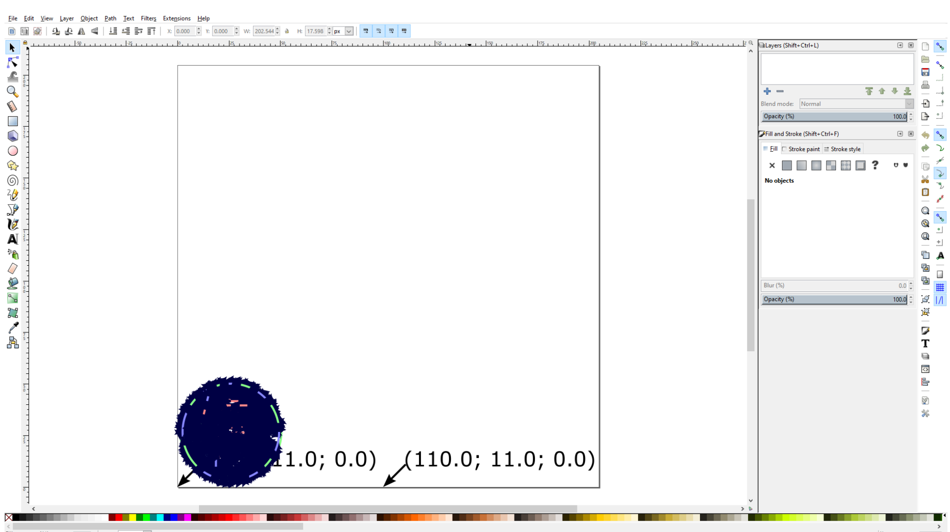

Figure 17: Laser tool extension for Inkscape.

5.6 Pronterface Graphical Interface

The **Pronterface** tool was used, which helps us manipulate our CNC machine in real time, usually used in 3D printer applications, since it has extruder calibration and other tools that conventional CNC does not have.

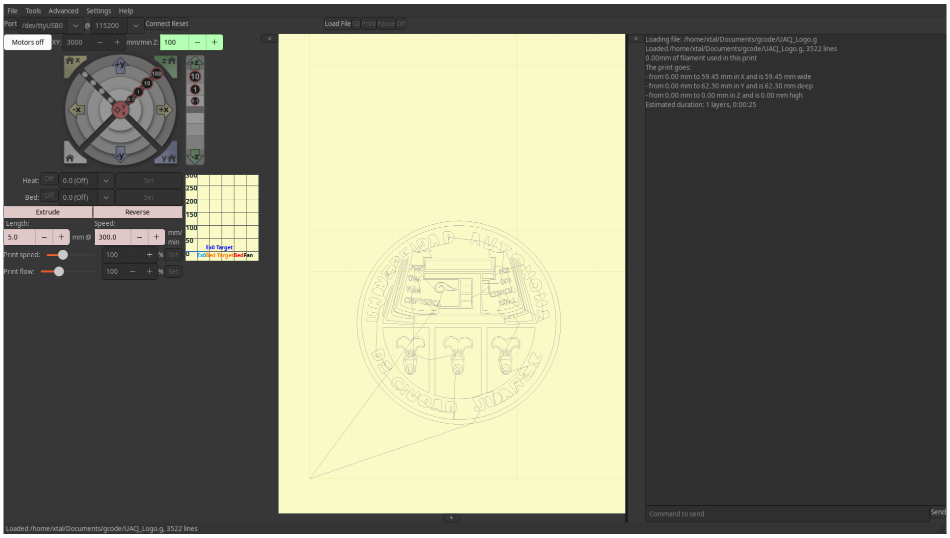

Figure 18: Pronterface user interface.

6 Results

In this project we were asked to build a machine capable of drawing our university's logo on a sheet of paper. At first, we faced several problems related to the structure, motor calibration, Marlin firmware adjustment, and converting the logo design to G-code, so that the machine could interpret it correctly.

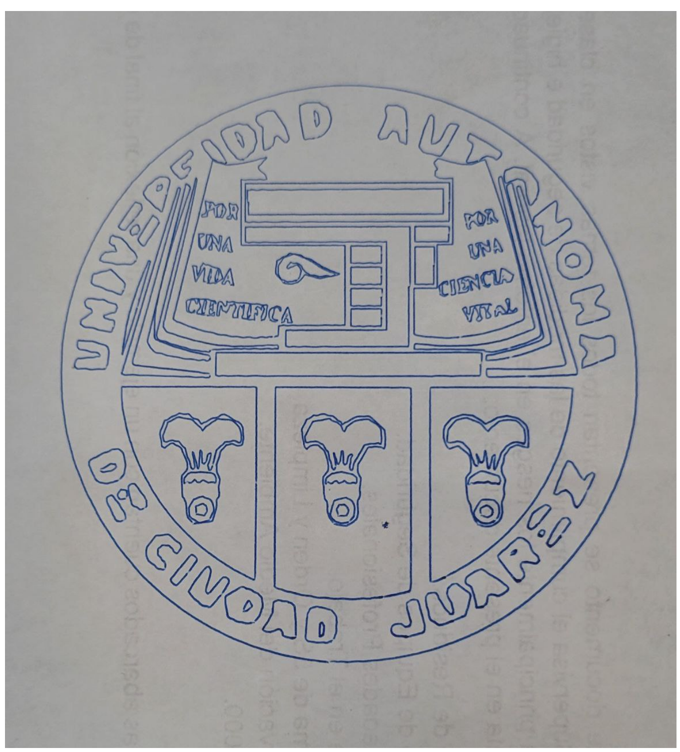

Figure 19: Results.

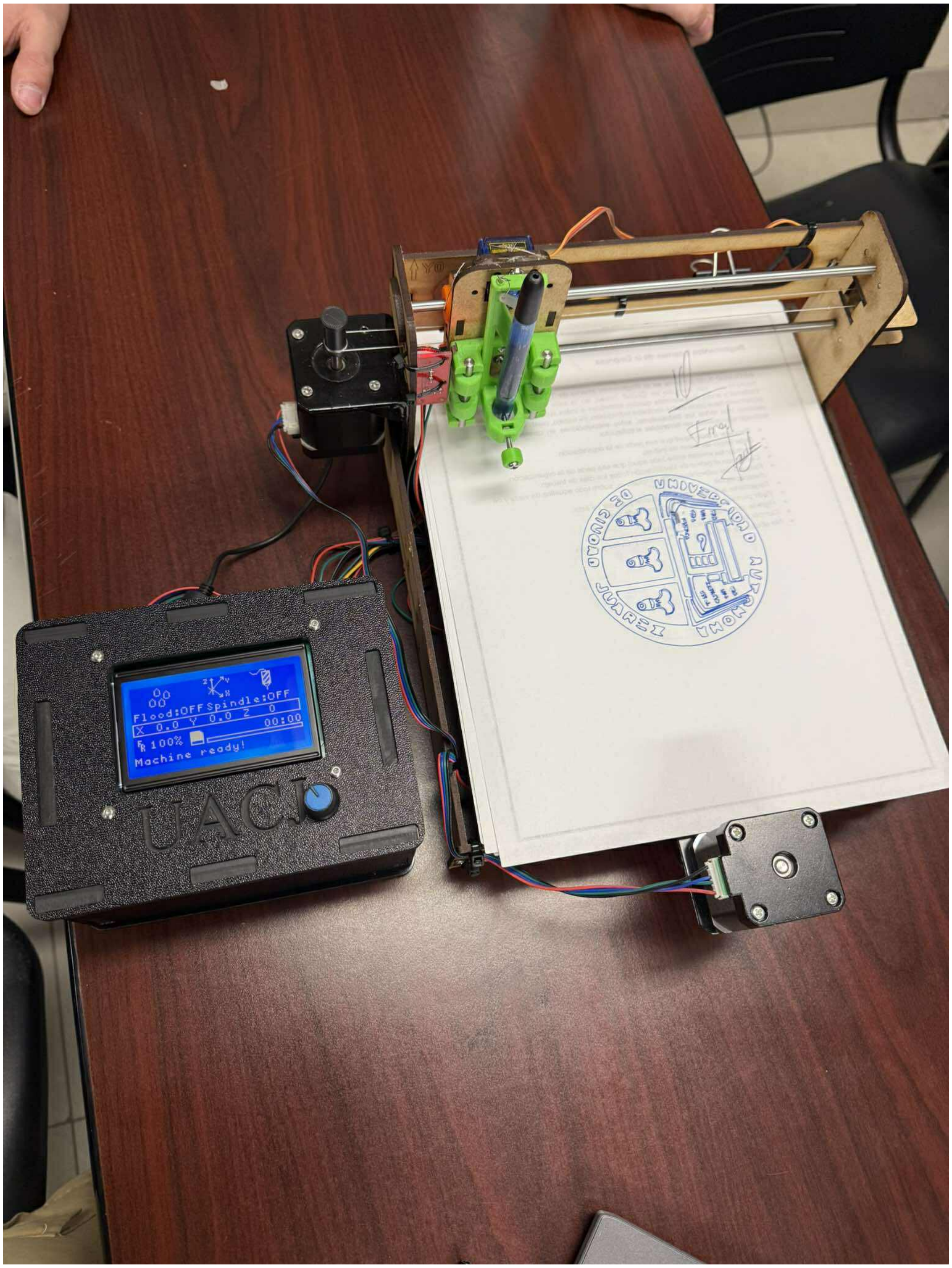

Figure 20: Results.

7 Conclusion

A favorable results were obtained, as the main objective of the project was met: tracing our university's logo on a sheet of paper using a CNC machine. To achieve this, it was necessary to correctly calibrate the NEMA 17 motors, adjust the firmware, precisely define the auto-homing point, and configure the appropriate servomotor angle.

8 Appendices

8.1 Code

8.1.1 UACJ_Logo.g

```
M280 P0 S20

G28 X Y
G90
G21
G1 F6000
G1  X34.7377 Y14.3117
G4 P0
 M280 P0 S120
G4 P0
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G2 X25.5981 Y27.4383 I-1.4462 J0.5988
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G3 X25.5268 Y26.8925 I1.2337 J0.0415
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G2 X23.6592 Y25.3257 I-0.8723 J1.0397
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G4 P0
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G2 X24.8044 Y26.4522 I-0.5966 J-0.3342
G2 X24.4796 Y25.9738 I-0.5115 J-0.0023
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G1 F6000
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G4 P0
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G4 P0
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G3 X25.0475 Y27.2482 I-1.0597 J0.0583
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G3 X23.576 Y27.2931 I-0.0344 J-1.1231
G3 X23.211 Y26.6299 I0.677 J-0.8046
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G2 X36.0279 Y26.0171 I-0.5489 J0.
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G1 F6000
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G4 P0
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G1  X51.2949 Y41.2486
G2 X49.4289 Y40.5422 I-9.8638 J23.2373
G2 X46.2399 Y39.8979 I-4.2006 J12.5758
G2 X42.2732 Y39.9946 I-1.5049 J19.6658
G3 X41.661 Y40.0494 I-1.0651 J-8.4454
G1  X41.5904 Y40.0316
G1  X41.5496 Y39.977
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G3 X48.2658 Y39.4483 I-0.5666 J4.9272
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G1 F6000
G1  X46.9872 Y38.3866
G4 P0
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G4 P0
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G1  X36.1447 Y39.3384
G1  X25.3022 Y39.3384
G1  X25.3022 Y38.3866
G1  X25.3022 Y37.4348
G1  X36.1447 Y37.4348
G1  X46.9872 Y37.4348
G1  X46.9872 Y38.3866
G4 P0
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G1 F6000
G1  X45.2451 Y40.5826
G4 P0
 M280 P0 S120
G4 P0
G1 F6000.000000
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G3 X48.6982 Y41.2838 I-2.7829 J13.7025
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G2 X46.2372 Y54.6196 I-0.599 J1.6428
G2 X45.5438 Y54.6226 I-0.3388 J1.868
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G3 X43.511 Y54.2581 I1.5569 J0.5804
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G1  X44.7525 Y50.8018
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G1  X44.0904 Y48.1119
G1  X43.4283 Y48.1119
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G1  X43.4283 Y44.5529
G1  X42.4749 Y44.5529
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G1  X41.5438 Y42.6751
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G3 X44.0904 Y40.5096 I3.8353 J39.1906
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G4 P0
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G1  X46.7178 Y43.194
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G1  X47.7811 Y43.1385
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G1  X46.8455 Y42.3044
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G1  X45.9941 Y41.8885
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G4 P0
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G4 P0
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G1 F6000
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G1  X48.3116 Y47.9472
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G4 P0
 M280 P0 S120
G4 P0
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G1  X47.1033 Y49.8312
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G1 F6000
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G4 P0
 M280 P0 S120
G4 P0
G1 F6000.000000
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G1  X46.0958 Y50.086
G1  X46.1381 Y50.1755
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M280 P0 S20
G1 F6000
G1  X45.796 Y49.8252
G4 P0
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G4 P0
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G4 P0
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G1 F6000
G1  X47.8149 Y50.721
G4 P0
 M280 P0 S120
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G4 P0
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G1 F6000
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G4 P0
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G1  X53.2919 Y48.5763
G4 P0
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 M280 P0 S120
G4 P0
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G2 X54.9756 Y47.034 I0.0392 J-0.5377
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M280 P0 S20
G1 F6000
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G4 P0
 M280 P0 S120
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M280 P0 S20
G1 F6000
G1  X56.133 Y41.8099
G4 P0
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G4 P0
G1 F6000.000000
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G1  X54.462 Y43.2245
G3 X54.7674 Y43.2719 I0.0906 J0.4234
G2 X55.0627 Y43.3043 I0.185 J-0.3236
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G3 X55.1707 Y42.9476 I0.361 J-0.0722
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G2 X54.9894 Y42.4912 I-0.4384 J0.3326
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G3 X54.6725 Y41.8311 I0.5507 J-0.2644
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G1  X56.133 Y41.8099
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G1 F6000
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G4 P0
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G4 P0
 M280 P0 S120
G4 P0
G1 F6000.000000
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G2 X45.8995 Y45.5151 I1.9951 J-5.6083
G2 X46.1479 Y45.4976 I0.0924 J-0.4604
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G1  X44.5574 Y44.062
G1  X44.5575 Y44.0619
G4 P0
M280 P0 S20
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G4 P0
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G4 P0
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G3 X44.2367 Y45.1621 I-0.1225 J0.4217
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G2 X43.6399 Y44.0753 I0.011 J0.438
G1  X43.6399 Y44.0751
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M280 P0 S20
G1 F6000
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 M280 P0 S120
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G1 F6000
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G4 P0
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G1  X35.1929 Y41.1594
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G1  X31.0132 Y40.4972
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G1  X36.1033 Y39.8351
G1  X41.1935 Y39.8351
G1  X41.1935 Y44.7184
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M280 P0 S20
G1 F6000
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G4 P0
 M280 P0 S120
G4 P0
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G1  X36.3931 Y44.5549
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G1  X38.9588 Y43.8514
G4 P0
M280 P0 S20
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G1  X37.6759 Y41.4925
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G1  X38.9588 Y42.196
G4 P0
M280 P0 S20
G1 F6000
G1  X38.9587 Y45.5881
G4 P0
 M280 P0 S120
G4 P0
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G1 F6000
G1  X38.9587 Y47.3263
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G1  X38.9587 Y47.3263
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G1 F6000
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G4 P0
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G1  X45.9819 Y47.9472
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G2 X46.2046 Y48.1336 I0.35 J-0.418
G1  X46.3436 Y48.1126
G1  X46.3684 Y48.0463
G3 X46.345 Y47.9155 I0.3303 J-0.1268
G3 X46.3739 Y47.5536 I2.6544 J0.0304
G3 X46.562 Y47.176 I0.6769 J0.1015
G1  X46.745 Y47.1751
G3 X46.8907 Y47.6461 I-0.3737 J0.3737
G3 X46.5688 Y48.1154 I-0.692 J-0.1297
G1  X46.5672 Y48.1653
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G2 X47.0835 Y48.2665 I0.1446 J-0.5246
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G1  X47.1666 Y47.1783
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G4 P0
 M280 P0 S120
G4 P0
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G3 X48.6893 Y55.3567 I-0.6502 J0.6916
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G3 X47.4322 Y56.4399 I-11.5356 J-15.6521
G3 X46.1623 Y57.1272 I-6.1504 J-9.8473
G1  X46.0251 Y57.0918
G3 X45.9146 Y56.7079 I0.4689 J-0.3429
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G2 X46.2074 Y56.4465 I-0.162 J-0.5103
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G2 X46.1637 Y55.5507 I-2.5102 J0.6406
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M280 P0 S20
G1 F6000
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G2 X45.0595 Y57.4397 I18.6384 J-10.2679
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G3 X42.3391 Y57.4234 I6.8622 J-3.4605
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G3 X44.3781 Y56.2026 I-4.4723 J3.5137
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G4 P0
M280 P0 S20
G1 F6000
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G4 P0
 M280 P0 S120
G4 P0
G1 F6000.000000
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G3 X40.7276 Y59.2218 I-0.3578 J-0.0589
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M280 P0 S20
G1 F6000
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G4 P0
 M280 P0 S120
G4 P0
G1 F6000.000000
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G4 P0
 M280 P0 S120
G4 P0
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G3 X33.7052 Y59.7465 I-1.0449 J-0.8505
G3 X32.8201 Y59.7671 I-0.5062 J-2.7137
G3 X32.1683 Y59.5323 I0.2301 J-1.6605
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G3 X32.2922 Y58.1995 I15.678 J1.4568
G3 X32.5207 Y56.929 I118.7551 J20.7024
G3 X32.5472 Y56.8239 I0.8337 J0.1546
G1  X32.6397 Y56.7577
G3 X33.3481 Y56.8373 I-0.062 J3.744
G1  X33.3481 Y56.8366
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G1 F6000
G1  X29.7716 Y56.0824
G4 P0
 M280 P0 S120
G4 P0
G1 F6000.000000
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G3 X30.1184 Y56.6114 I-0.388 J0.2544
G2 X30.1539 Y56.924 I0.8437 J0.0625
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G1  X30.461 Y57.0345
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G1  X31.3354 Y56.7825
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G3 X28.9219 Y57.9211 I0.8498 J0.3166
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G1  X29.1891 Y56.879
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G1  X29.3509 Y55.8961
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G1  X29.7716 Y56.0824
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G1 F6000
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G4 P0
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G3 X27.2254 Y58.0642 I-0.5715 J-1.1055
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G3 X25.8077 Y57.5057 I3.0693 J-5.6315
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G3 X27.5783 Y55.1266 I0.2647 J0.3374
G1  X27.5783 Y55.1266
G4 P0
M280 P0 S20
G1 F6000
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G4 P0
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G4 P0
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G3 X24.3462 Y56.6962 I-0.1894 J-0.2921
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G1  X24.5096 Y55.8748
G3 X24.9488 Y55.3032 I2.8142 J1.7086
G3 X25.3209 Y55.0297 I0.9747 J0.9358
G3 X26.1106 Y54.7701 I1.2847 J2.5778
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G1  X26.1864 Y54.8786
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G1 F6000
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G4 P0
 M280 P0 S120
G4 P0
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G1  X27.4956 Y56.1192
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G1 F6000
G1  X22.9953 Y53.8441
G4 P0
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G1  X22.5124 Y54.55
G2 X22.6749 Y54.7955 I0.3885 J-0.0806
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G3 X23.0752 Y55.7989 I0.1366 J-0.3544
G3 X22.3275 Y55.1797 I16.5927 J-20.7965
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G1  X21.7224 Y54.329
G3 X22.2523 Y53.6904 I2.5392 J1.5678
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G4 P0
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G4 P0
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G1  X21.4182 Y53.5868
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G3 X20.981 Y51.0795 I-1.8019 J0.6664
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G1 F6000
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G4 P0
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G4 P0
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G1  X20.9467 Y52.9965
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G4 P0
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G4 P0
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G4 P0
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G4 P0
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G4 P0
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G3 X19.6736 Y40.2405 I0.5282 J-0.1288
G3 X20.2846 Y39.4129 I0.8701 J0.003
G3 X24.6608 Y38.6849 I4.8184 J15.4477
G3 X24.8995 Y38.7499 I0.0123 J0.4255
G1  X24.9712 Y38.8795
G1  X24.9712 Y38.8795
G4 P0
M280 P0 S20
G1 F6000
G1  X24.9712 Y39.5041
G4 P0
 M280 P0 S120
G4 P0
G1 F6000.000000
G1  X25.0515 Y39.5893 F6000.000000
G2 X27.7439 Y39.6696 I2.6924 J-45.1099
G1  X30.5166 Y39.6696
G1  X30.5166 Y39.8466
G1  X30.4395 Y39.9387
G3 X29.8751 Y39.962 I-0.3667 J-2.0374
G2 X26.7371 Y39.8496 I-2.5065 J26.1163
G2 X24.609 Y40.0889 I0.2935 J12.1913
G2 X21.2558 Y41.0305 I4.6761 J23.094
G2 X20.9983 Y41.3976 I0.1331 J0.3671
G2 X21.0082 Y41.4793 I0.3444 J-0.
G2 X22.489 Y47.5015 I1903.913 J-464.9596
G3 X23.8268 Y53.5179 I-125.207 J30.9976
G1  X23.6992 Y53.6322
G1  X23.6654 Y53.5965
G3 X22.0469 Y47.2761 I1698.7824 J-438.4048
G3 X20.462 Y40.9908 I1708.8132 J-434.222
G3 X20.4223 Y40.7362 I1.3862 J-0.3468
G1  X20.6179 Y40.4353
G3 X23.763 Y39.5042 I8.6655 J23.4934
G3 X24.8498 Y39.3829 I1.0864 J4.8064
G1  X24.9712 Y39.5043
G1  X24.9712 Y39.5041
G4 P0
M280 P0 S20
G1 F6000
G1  X24.5987 Y42.3723
G4 P0
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G4 P0
G1 F6000.000000
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G1  X23.7478 Y42.5747
G3 X24.0984 Y43.0847 I-0.3414 J0.6101
G3 X23.9055 Y43.6649 I-0.6753 J0.0976
G1  X23.8528 Y43.6605
G1  X23.8124 Y43.5345
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G1  X23.6469 Y43.4251
G1  X23.6192 Y43.5259
G3 X23.5132 Y43.65 I-0.4126 J-0.2449
G1  X23.2574 Y43.6155
G3 X23.0675 Y43.0122 I0.8631 J-0.6033
G3 X23.1393 Y42.7207 I0.6281 J0.
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G3 X23.436 Y42.6787 I-0. J0.5484
G1  X23.5476 Y42.7482
G1  X23.619 Y42.7798
G1  X23.6469 Y42.7532
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G2 X23.1306 Y43.7168 I0.4338 J-0.2243
G2 X23.6932 Y43.7807 I0.4163 J-1.1569
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G3 X24.7022 Y43.6821 I5.8736 J59.7392
G2 X25.3019 Y43.5464 I-0.1966 J-2.2615
G1  X25.385 Y43.4265
G1  X25.3006 Y43.3273
G1  X25.1955 Y43.3957
G1  X25.0148 Y43.5891
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G2 X24.8805 Y43.4378 I-0.3433 J0.0066
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G1  X25.1094 Y43.1474
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G3 X25.0498 Y42.775 I1.3568 J-0.217
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G1  X24.9887 Y42.5141
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G1  X25.2749 Y42.6221
G3 X25.2284 Y42.4689 I1.3379 J-0.4897
G1  X25.2195 Y42.3945
G1  X25.1752 Y42.3487
G2 X24.5987 Y42.3728 I-0.1311 J3.7678
G1  X24.5987 Y42.3723
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G1 F6000
G1  X24.4244 Y42.6146
G4 P0
 M280 P0 S120
G4 P0
G1 F6000.000000
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G1  X24.4839 Y43.6421
G1  X24.3672 Y43.5623
G3 X24.1825 Y42.869 I3.006 J-1.1719
G3 X24.1863 Y42.5365 I1.0301 J-0.1545
G1  X24.2494 Y42.4834
G1  X24.3509 Y42.5173
G1  X24.4244 Y42.6145
G1  X24.4244 Y42.6146
G4 P0
M280 P0 S20
G1 F6000
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G1  X24.3376 Y46.2521
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G2 X23.5242 Y45.6906 I5.7471 J1.1509
G3 X23.4362 Y46.2372 I-5.0731 J-0.5371
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G2 X23.6055 Y46.4562 I0.2121 J-0.4612
G2 X23.806 Y46.4047 I-0. J-0.4163
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G1  X24.3009 Y46.6426
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G1 F6000
G1  X24.9643 Y45.5499
G4 P0
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G4 P0
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G4 P0
M280 P0 S20
G1 F6000
G1  X25.5424 Y45.3927
G4 P0
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G4 P0
G1 F6000.000000
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G3 X25.2616 Y45.5666 I1.2511 J-0.6756
G3 X25.2653 Y45.1857 I1.3385 J-0.1774
G1  X25.3278 Y45.132
G1  X25.4205 Y45.1758
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G1  X25.5424 Y45.3927
G4 P0
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G4 P0
G1 F6000.000000
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G1  X26.3198 Y45.6187
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G1  X26.3368 Y45.3802
G1  X26.4015 Y45.4176
G1  X26.4081 Y45.5044
G1  X26.4081 Y45.5044
G4 P0
M280 P0 S20
G1 F6000
G1  X26.5734 Y44.8581
G4 P0
 M280 P0 S120
G4 P0
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G1  X25.8309 Y45.0259
G1  X25.8607 Y45.0492
G1  X25.9137 Y45.079
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G2 X26.5068 Y46.1859 I8.0718 J-3.9046
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G1  X26.5763 Y46.1836
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G1  X26.8216 Y44.8121
G2 X26.7508 Y44.8242 I0.3339 J2.1713
G2 X26.5733 Y44.8579 I6.3947 J34.1229
G1  X26.5734 Y44.8581
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G1 F6000
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G4 P0
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G4 P0
G1 F6000.000000
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G1  X26.0727 Y47.9239
G1  X25.9946 Y47.7385
G1  X25.9802 Y47.6576
G1  X25.8866 Y47.6151
G1  X25.8386 Y47.6791
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G1  X26.4624 Y48.8144
G1  X26.4988 Y48.7626
G2 X26.5806 Y48.2592 I-5.1338 J-1.0935
G3 X26.668 Y47.7531 I4.3874 J0.4965
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M280 P0 S20
G1 F6000
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G4 P0
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G4 P0
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G1  X26.1299 Y48.1066
G1  X26.1534 Y48.0523
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G1  X26.2662 Y48.058
G1  X26.2955 Y48.1579
G1  X26.2955 Y48.1579
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M280 P0 S20
G1 F6000
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G4 P0
 M280 P0 S120
G4 P0
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G3 X24.9453 Y48.9103 I-0.5002 J0.0887
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G3 X24.6829 Y48.4647 I1.0355 J-0.6099
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G2 X24.1835 Y47.7809 I-0.276 J0.2016
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G2 X24.3569 Y49.0705 I0. J-0.7031
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G1  X24.3694 Y48.9621
G1  X24.3204 Y48.9396
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G3 X24.5841 Y48.614 I-2.7062 J0.4974
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G1  X24.9138 Y49.0219
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G1  X25.741 Y48.9397
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G1  X25.3038 Y48.0844
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G4 P0
 M280 P0 S120
G4 P0
G1 F6000.000000
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G1  X25.2926 Y50.439
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M280 P0 S20
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G4 P0
M280 P0 S20
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G1  X30.8402 Y45.0111
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G4 P0
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G3 X32.3375 Y46.0111 I-1.0059 J0.5068
G3 X31.9886 Y46.4812 I-0.4912 J0.
G3 X31.4139 Y46.2906 I-0.1522 J-0.5028
G2 X31.1944 Y46.1593 I-0.2673 J0.1976
G1  X31.0959 Y46.2445
G2 X31.4236 Y46.6868 I0.4624 J0.
G2 X32.1622 Y46.5694 I0.2489 J-0.8167
G2 X32.3258 Y46.411 I-0.4128 J-0.5897
G2 X32.4708 Y46.1572 I-0.9513 J-0.7118
G3 X32.8006 Y45.7877 I0.699 J0.2921
G3 X33.5286 Y45.5668 I0.8711 J1.5608
G3 X35.0312 Y45.5667 I0.7525 J9.3766
G1  X35.1514 Y45.6971
G1  X35.1264 Y45.7343
G3 X34.6581 Y45.8976 I-1.8647 J-4.5933
G2 X34.187 Y46.0882 I0.7289 J2.4797
G2 X33.5202 Y46.4942 I2.8284 J5.3949
G3 X32.2174 Y47.2739 I-6.0711 J-8.6661
G3 X31.7668 Y47.3794 I-0.443 J-0.8775
G3 X30.844 Y47.0795 I0.013 J-1.6099
G3 X30.3977 Y46.401 I0.655 J-0.9168
G3 X30.4119 Y45.7795 I1.2986 J-0.2811
G3 X30.6784 Y45.3422 I0.871 J0.2309
G3 X31.2472 Y45.0932 I0.6563 J0.7252
G3 X31.9434 Y45.2285 I0.1142 J1.2718
G1  X31.9434 Y45.2286
G4 P0
M280 P0 S20
G1 F6000
G1  X31.1787 Y42.3595
G4 P0
 M280 P0 S120
G4 P0
G1 F6000.000000
G1  X31.1498 Y42.4548 F6000.000000
G1  X31.0959 Y42.4837
G1  X31.042 Y42.4548
G1  X31.0131 Y42.3595
```

```
G1  X31.042 Y42.2642
G1  X31.0959 Y42.2354
G1  X31.1498 Y42.2642
G1  X31.1787 Y42.3595
G1  X31.1787 Y42.3595
G4 P0
M280 P0 S20
G1 F6000
G1  X30.7063 Y41.7924
G4 P0
 M280 P0 S120
G4 P0
G1 F6000.000000
G1  X30.6748 Y41.859 F6000.000000
G2 X30.9107 Y42.448 I3.8597 J-1.2035
G2 X31.2283 Y43.0416 I9.5019 J-4.7019
G1  X31.2653 Y43.063
G1  X31.2961 Y43.0396
G2 X31.4182 Y42.5062 I-6.1595 J-1.6909
G3 X31.5463 Y41.9612 I5.6695 J1.0449
G3 X31.6082 Y41.844 I0.3104 J0.089
G1  X31.5947 Y41.7863
G2 X31.3751 Y41.7399 I-0.2178 J0.4881
G2 X31.1743 Y41.7757 I0.0021 J0.5935
G1  X31.1701 Y41.8145
G1  X31.2369 Y42.0287
G1  X31.0341 Y42.1347
G1  X30.9401 Y42.0672
G1  X30.9087 Y41.9217
G1  X30.8531 Y41.8059
G1  X30.7063 Y41.7926
G1  X30.7063 Y41.7924
G4 P0
M280 P0 S20
G1 F6000
G1  X29.762 Y41.9967
G4 P0
 M280 P0 S120
G4 P0
G1 F6000.000000
G2 X29.6506 Y42.1934 I0.2995 J0.2996
G2 X29.6327 Y42.5131 I0.8251 J0.2067
G2 X29.8839 Y42.9704 I0.7168 J-0.0962
G2 X30.3616 Y43.1263 I0.4319 J-0.5134
G2 X30.616 Y43.0427 I-0.0388 J-0.5473
G1  X30.682 Y42.9231
G2 X30.6535 Y42.7577 I-0.4931 J-0.0003
G1  X30.5993 Y42.6906
G1  X30.5472 Y42.694
G1  X30.5165 Y42.7584
G1  X30.4881 Y42.8638
G3 X30.3829 Y42.9882 I-0.4032 J-0.2344
G1  X30.2576 Y43.0147
G3 X30.0932 Y42.9063 I0.0888 J-0.3135
G3 X29.9043 Y42.1488 I0.718 J-0.5814
G1  X30.271 Y41.9905
G1  X30.413 Y42.0116
G1  X30.4338 Y41.9455
G2 X30.1423 Y41.8219 I-0.2654 J0.2203
G2 X29.762 Y41.9965 I0.0455 J0.6004
G1  X29.762 Y41.9967
G4 P0
M280 P0 S20
G1 F6000
G1  X29.044 Y41.9604
G4 P0
 M280 P0 S120
G4 P0
G1 F6000.000000
G1  X28.9946 Y42.0186 F6000.000000
G1  X29.0131 Y42.0986
```

```
G3 X29.338 Y42.9005 I-1.1934 J0.9503
G1  X29.1155 Y43.1458
G1  X29.0599 Y43.1117
G3 X29.0212 Y42.9596 I0.3217 J-0.163
G1  X29.0155 Y42.7734
G1  X28.9256 Y42.9596
G1  X28.6875 Y43.0923
G1  X28.5327 Y42.9183
G1  X28.583 Y42.8351
G1  X28.6569 Y42.8512
G1  X28.7279 Y42.8627
G1  X28.7771 Y42.7684
G2 X28.7507 Y42.5325 I-1.1145 J0.0055
G2 X28.6956 Y42.4008 I-0.3943 J0.0875
G1  X28.6457 Y42.3811
G1  X28.6141 Y42.4213
G1  X28.5806 Y42.5337
G1  X28.521 Y42.5661
G1  X28.4731 Y42.5271
G3 X28.4624 Y42.2812 I0.7589 J-0.1561
G1  X28.4957 Y41.9962
G1  X27.9542 Y42.0443
G2 X27.4426 Y42.1534 I0.1921 J2.155
G1  X27.4348 Y42.223
G3 X27.5997 Y42.4456 I-0.1964 J0.3178
G3 X27.7024 Y43.2394 I-3.0153 J0.7938
G1  X27.6268 Y43.2833
G1  X27.5009 Y43.1245
G1  X27.4456 Y42.9382
G1  X27.3653 Y43.1365
G1  X27.2705 Y43.1468
G3 X27.1163 Y42.7743 I0.7978 J-0.5485
G3 X27.0833 Y42.3181 I2.04 J-0.377
G1  X27.1373 Y42.2241
G1  X27.1331 Y42.1872
G2 X26.9368 Y42.1531 I-0.1987 J0.5614
G2 X26.7076 Y42.2407 I-0.0009 J0.3412
G1  X26.7041 Y42.3629
G3 X27.0341 Y43.0573 I-0.8299 J0.82
G1  X26.8067 Y43.3106
G1  X26.7328 Y43.2838
G3 X26.6331 Y43.1657 I0.2858 J-0.3426
G1  X26.583 Y43.1515
G1  X26.5462 Y43.212
G1  X26.467 Y43.2335
G3 X26.343 Y42.8765 I0.7755 J-0.4695
G2 X26.1378 Y42.604 I-0.3399 J0.0425
G1  X25.9107 Y42.6915
G1  X25.6834 Y43.1864
G1  X25.6146 Y42.8589
G3 X25.5703 Y42.5333 I2.2469 J-0.4715
G1  X25.587 Y42.4244
G1  X25.5692 Y42.3578
G1  X25.455 Y42.3173
G1  X25.3773 Y42.3621
G1  X25.3839 Y42.4403
G1  X25.448 Y42.5774
G3 X25.4975 Y43.0604 I-3.026 J0.5543
G2 X25.5793 Y43.486 I1.308 J-0.0305
G1  X25.6827 Y43.5582
G1  X25.8193 Y43.5012
G2 X26.0047 Y43.2301 I-0.669 J-0.6564
G1  X26.175 Y43.1334
G1  X26.2714 Y43.2609
G1  X26.2291 Y43.5211
G1  X27.8969 Y43.3641
G2 X29.5594 Y43.1949 I-10.4107 J-110.5263
G1  X29.5972 Y43.1781
G1  X29.6042 Y43.148
G2 X29.5362 Y43.0364 I-0.4166 J0.1777
G1  X29.4772 Y42.9105
```

```
G3 X29.421 Y42.4144 I3.5041 J-0.6489
G2 X29.2677 Y42.0163 I-0.668 J0.0286
G1  X29.0439 Y41.9594
G1  X29.044 Y41.9604
G4 P0
M280 P0 S20
G1 F6000
G1  X28.1455 Y42.3803
G4 P0
 M280 P0 S120
G4 P0
G1 F6000.000000
G3 X28.2278 Y43.1657 I-4.8691 J0.9067
G1  X28.1663 Y43.2286
G1  X28.0433 Y43.1481
G3 X27.9 Y42.6191 I1.71 J-0.7469
G3 X27.9084 Y42.2088 I1.489 J-0.1749
G1  X27.9742 Y42.1526
G1  X28.0653 Y42.1981
G3 X28.1455 Y42.3803 I-0.3478 J0.2619
G1  X28.1455 Y42.3803
G4 P0
M280 P0 S20
G1 F6000
G1  X29.9786 Y40.6503
G4 P0
 M280 P0 S120
G4 P0
G1 F6000.000000
G1  X30.4752 Y40.7042
G1  X30.5002 Y41.1387
G1  X30.5253 Y41.5733
G1  X33.2532 Y41.5733
G1  X35.9812 Y41.5733
G1  X35.9595 Y44.8219
G1  X35.9379 Y48.0705
G1  X31.6547 Y48.0918
G1  X27.3715 Y48.1131
G1  X27.3715 Y50.8023
G1  X27.3715 Y53.4915
G1  X28.4503 Y53.4915
G1  X29.5292 Y53.4915
G1  X29.4322 Y53.6777
G3 X29.3149 Y53.8474 I-0.7393 J-0.3859
G3 X29.0569 Y54.0976 I-1.8704 J-1.6708
G2 X28.8096 Y54.3454 I1.2764 J1.521
G1  X28.7802 Y54.4286
G1  X28.8108 Y54.5136
G2 X29.1513 Y54.893 I3.4703 J-2.7724
G3 X29.4939 Y55.2599 I-4.6869 J4.7193
G1  X29.4929 Y55.2877
G1  X29.4602 Y55.2965
G3 X28.4443 Y55.0218 I8.0921 J-31.934
G2 X26.2336 Y54.6704 I-2.5309 J8.7934
G2 X25.3213 Y54.9124 I-0.0569 J1.6259
G3 X25.0492 Y55.0594 I-1.398 J-2.2625
G1  X25.0208 Y55.0529
G1  X25.0037 Y55.0182
G3 X23.3779 Y48.5378 I2605.449 J-657.0854
G1  X21.7596 Y42.0529
G3 X25.9032 Y40.7706 I6.7239 J14.3924
G3 X29.9786 Y40.6503 I2.4852 J15.097
G1  X29.9786 Y40.6503
G4 P0
M280 P0 S20
G1 F6000
G1  X44.1375 Y21.2071
G4 P0
 M280 P0 S120
G4 P0
G1 F6000.000000
```

```
G3 X49.1208 Y24.8108 I-8.377 J16.8312
G3 X52.6242 Y29.779 I-13.0555 J12.9257
G3 X53.8582 Y33.074 I-17.7151 J8.5132
G3 X54.3679 Y36.1107 I-14.5138 J3.9969
G1  X54.3917 Y36.4832
G1  X48.5789 Y36.5044
G1  X42.7661 Y36.5256
G1  X42.7661 Y28.5794
G3 X42.8203 Y20.6874 I574.337 J0.
G1  X42.8753 Y20.6331
G1  X42.9792 Y20.6556
G3 X44.1375 Y21.207 I-15.1502 J33.3205
G1  X44.1375 Y21.2071
G4 P0
M280 P0 S20
G1 F6000
G1 X0 Y0
G28 X0 Y0
M18
```

8.1.2 Configuration.h

```
#ifndef CONFIGURATION_H
#define CONFIGURATION_H
#define CONFIGURATION_H_VERSION 010100

#define STRING_CONFIG_H_AUTHOR "(none, default config)"
#define SHOW_BOOTSCREEN
#define STRING_SPLASH_LINE1 SHORT_BUILD_VERSION
#define STRING_SPLASH_LINE2 WEBSITE_URL

#define SERIAL_PORT 0
#define BAUDRATE 115200

#ifndef MOTHERBOARD
  #define MOTHERBOARD BOARD_RAMPS_14_EFF
#endif

#define EXTRUDERS 2

#define TEMP_SENSOR_0 998
#define TEMP_SENSOR_1 998
#define TEMP_SENSOR_2 0
#define TEMP_SENSOR_3 0
#define TEMP_SENSOR_4 0
#define TEMP_SENSOR_BED 0

#define DUMMY_THERMISTOR_998_VALUE 25
#define DUMMY_THERMISTOR_999_VALUE 100

#define TEMP_RESIDENCY_TIME 10
#define TEMP_HYSTERESIS 3
#define TEMP_WINDOW     1

#define TEMP_BED_RESIDENCY_TIME 10
#define TEMP_BED_HYSTERESIS 3
#define TEMP_BED_WINDOW     1

#define HEATER_0_MINTEMP 5
#define HEATER_1_MINTEMP 5
#define HEATER_2_MINTEMP 5
#define HEATER_3_MINTEMP 5
#define HEATER_4_MINTEMP 5
#define BED_MINTEMP 5

#define HEATER_0_MAXTEMP 275
#define HEATER_1_MAXTEMP 275
#define HEATER_2_MAXTEMP 275
#define HEATER_3_MAXTEMP 275
#define HEATER_4_MAXTEMP 275
#define BED_MAXTEMP 150

#define PIDTEMP
#define BANG_MAX 255
#define PID_MAX BANG_MAX
#if ENABLED(PIDTEMP)
  #define PID_FUNCTIONAL_RANGE 10
  #define K1 0.95
  #define DEFAULT_Kp 22.2
  #define DEFAULT_Ki 1.08
  #define DEFAULT_Kd 114
#endif

#define MAX_BED_POWER 255

#define PREVENT_COLD_EXTRUSION
#define EXTRUDE_MINTEMP 170
#define PREVENT_LENGTHY_EXTRUDE
#define EXTRUDE_MAXLENGTH 200

#define THERMAL_PROTECTION_HOTENDS
```

```
#define THERMAL_PROTECTION_BED

#define USE_XMIN_PLUG
#define USE_YMIN_PLUG
#define USE_ZMIN_PLUG
#define USE_ZMAX_PLUG

#define ENDSTOPPULLUPS
#if DISABLED(ENDSTOPPULLUPS)
  #define ENDSTOPPULLUP_XMIN
  #define ENDSTOPPULLUP_YMIN
  #define ENDSTOPPULLUP_ZMIN
  #define ENDSTOPPULLUP_ZMIN_PROBE
#endif

#define X_MIN_ENDSTOP_INVERTING true
#define Y_MIN_ENDSTOP_INVERTING true
#define Z_MIN_ENDSTOP_INVERTING true
#define X_MAX_ENDSTOP_INVERTING true
#define Y_MAX_ENDSTOP_INVERTING true
#define Z_MAX_ENDSTOP_INVERTING true
#define Z_MIN_PROBE_ENDSTOP_INVERTING false

#define DEFAULT_AXIS_STEPS_PER_UNIT   { 400, 400, 400, 500 }
#define DEFAULT_MAX_FEEDRATE          { 5000, 5000, 5000, 25 }
#define DEFAULT_MAX_ACCELERATION      { 500, 500, 100, 10000 }
#define DEFAULT_ACCELERATION          300
#define DEFAULT_RETRACT_ACCELERATION  800
#define DEFAULT_TRAVEL_ACCELERATION   200
#define DEFAULT_XJERK                 5.0
#define DEFAULT_YJERK                 5.0
#define DEFAULT_ZJERK                  0.4
#define DEFAULT_EJERK                  5.0

#define Z_MIN_PROBE_USES_Z_MIN_ENDSTOP_PIN

#define X_PROBE_OFFSET_FROM_EXTRUDER 10
#define Y_PROBE_OFFSET_FROM_EXTRUDER 10
#define Z_PROBE_OFFSET_FROM_EXTRUDER 0
#define XY_PROBE_SPEED 8000
#define Z_PROBE_SPEED_FAST HOMING_FEEDRATE_Z
#define Z_PROBE_SPEED_SLOW (Z_PROBE_SPEED_FAST / 2)
#define Z_CLEARANCE_DEPLOY_PROBE   10
#define Z_CLEARANCE_BETWEEN_PROBES  5
#define Z_PROBE_OFFSET_RANGE_MIN -20
#define Z_PROBE_OFFSET_RANGE_MAX 20

#define X_ENABLE_ON 0
#define Y_ENABLE_ON 0
#define Z_ENABLE_ON 0
#define E_ENABLE_ON 0

#define DISABLE_X false
#define DISABLE_Y false
#define DISABLE_Z false
#define DISABLE_REDUCED_ACCURACY_WARNING

#define DISABLE_E false
#define DISABLE_INACTIVE_EXTRUDER true

#define INVERT_X_DIR false
#define INVERT_Y_DIR true
#define INVERT_Z_DIR false

#define X_DRIVER_TYPE DRV8825
#define Y_DRIVER_TYPE DRV8825

#define INVERT_E0_DIR false
#define INVERT_E1_DIR false
#define INVERT_E2_DIR false
#define INVERT_E3_DIR false
```

```
#define INVERT_E4_DIR false

#define X_HOME_DIR -1
#define Y_HOME_DIR -1
#define Z_HOME_DIR 1

#define X_BED_SIZE 100
#define Y_BED_SIZE 80
#define X_MIN_POS 0
#define Y_MIN_POS 0
#define Z_MIN_POS 0
#define X_MAX_POS X_BED_SIZE
#define Y_MAX_POS Y_BED_SIZE
#define Z_MAX_POS 100

#define MANUAL_X_HOME_POS 0
#define MANUAL_Y_HOME_POS 0
#define MANUAL_Z_HOME_POS 0

#define HOMING_FEEDRATE_XY (10*60)
#define HOMING_FEEDRATE_Z  (5*60)

#define EEPROM_SETTINGS
#define DISABLE_M503
#define EEPROM_CHITCHAT

#define HOST_KEEPALIVE_FEATURE
#define DEFAULT_KEEPALIVE_INTERVAL 2

#define PREHEAT_1_TEMP_HOTEND 180
#define PREHEAT_1_TEMP_BED     70
#define PREHEAT_1_FAN_SPEED     0

#define PREHEAT_2_TEMP_HOTEND 240
#define PREHEAT_2_TEMP_BED    110
#define PREHEAT_2_FAN_SPEED     0

#define LCD_LANGUAGE en
#define DISPLAY_CHARSET_HD44780 JAPANESE
#define DOGLCD
#define SDSUPPORT
#define REVERSE_MENU_DIRECTION
#define INDIVIDUAL_AXIS_HOMING_MENU
#define SPEAKER
#define LCD_FEEDBACK_FREQUENCY_DURATION_MS 0
#define LCD_FEEDBACK_FREQUENCY_HZ 0
#define REPRAP_DISCOUNT_FULL_GRAPHIC_SMART_CONTROLLER

#define FAST_PWM_FAN

#define NUM_SERVOS 1
#define SERVO_DELAY 300

#define DEFAULT_NOMINAL_FILAMENT_DIA 1.7500

#endif
```

8.1.3 Configuration_adv.h

```
#ifndef CONFIGURATION_ADV_H
#define CONFIGURATION_ADV_H
#define CONFIGURATION_ADV_H_VERSION 010100

#if DISABLED(PIDTEMPBED)
  #define BED_CHECK_INTERVAL 5000
  #if ENABLED(BED_LIMIT_SWITCHING)
    #define BED_HYSTERESIS 2
  #endif
#endif

#if ENABLED(THERMAL_PROTECTION_HOTENDS)
  #define THERMAL_PROTECTION_PERIOD 40
  #define THERMAL_PROTECTION_HYSTERESIS 4
  #define WATCH_TEMP_PERIOD 20
  #define WATCH_TEMP_INCREASE 2
#endif

#if ENABLED(THERMAL_PROTECTION_BED)
  #define THERMAL_PROTECTION_BED_PERIOD 20
  #define THERMAL_PROTECTION_BED_HYSTERESIS 2
  #define WATCH_BED_TEMP_PERIOD 60
  #define WATCH_BED_TEMP_INCREASE 2
#endif

#if ENABLED(PIDTEMP)
  #if ENABLED(PID_EXTRUSION_SCALING)
    #define DEFAULT_Kc (100)
    #define LPQ_MAX_LEN 50
  #endif
#endif

#define AUTOTEMP
#if ENABLED(AUTOTEMP)
  #define AUTOTEMP_OLDWEIGHT 0.98
#endif

#define TEMP_SENSOR_AD595_OFFSET 0.0
#define TEMP_SENSOR_AD595_GAIN   1.0

#if ENABLED(USE_CONTROLLER_FAN)
  #define CONTROLLERFAN_SECS 60
  #define CONTROLLERFAN_SPEED 255
#endif

#define E0_AUTO_FAN_PIN -1
#define E1_AUTO_FAN_PIN -1
#define E2_AUTO_FAN_PIN -1
#define E3_AUTO_FAN_PIN -1
#define E4_AUTO_FAN_PIN -1
#define EXTRUDER_AUTO_FAN_TEMPERATURE 50
#define EXTRUDER_AUTO_FAN_SPEED   255

#define Z_PROBE_SERVO_NR 1

#if ENABLED(CASE_LIGHT_ENABLE)
  #define INVERT_CASE_LIGHT false
  #define CASE_LIGHT_DEFAULT_ON true
  #define CASE_LIGHT_DEFAULT_BRIGHTNESS 105
#endif

#define X_HOME_BUMP_MM 5
#define Y_HOME_BUMP_MM 5
#define Z_HOME_BUMP_MM 2
#define HOMING_BUMP_DIVISOR {2, 2, 4}

#define AXIS_RELATIVE_MODES {false, false, false, false}

#define INVERT_X_STEP_PIN false
```

```
#define INVERT_Y_STEP_PIN false
#define INVERT_Z_STEP_PIN false
#define INVERT_E_STEP_PIN false

#define DEFAULT_STEPPER_DEACTIVE_TIME 120
#define DISABLE_INACTIVE_X true
#define DISABLE_INACTIVE_Y true
#define DISABLE_INACTIVE_Z true
#define DISABLE_INACTIVE_E true

#define DEFAULT_MINIMUMFEEDRATE       0.0
#define DEFAULT_MINTRAVELFEEDRATE     0.0

#if ENABLED(ULTIPANEL)
  #define MANUAL_FEEDRATE {50*60, 50*60, 4*60, 60}
#endif

#define DEFAULT_MINSEGMENTTIME        20000

#define MINIMUM_PLANNER_SPEED 0.05
#define MICROSTEP_MODES {16,16,16,16,16}

#if ENABLED(HAVE_TMCDRIVER)
  #define X_MAX_CURRENT     1000
  #define X_SENSE_RESISTOR    91
  #define X_MICROSTEPS        16
  #define X2_MAX_CURRENT    1000
  #define X2_SENSE_RESISTOR   91
  #define X2_MICROSTEPS       16
  #define Y_MAX_CURRENT     1000
  #define Y_SENSE_RESISTOR    91
  #define Y_MICROSTEPS        16
  #define Y2_MAX_CURRENT    1000
  #define Y2_SENSE_RESISTOR   91
  #define Y2_MICROSTEPS       16
  #define Z_MAX_CURRENT     1000
  #define Z_SENSE_RESISTOR    91
  #define Z_MICROSTEPS        16
  #define Z2_MAX_CURRENT    1000
  #define Z2_SENSE_RESISTOR   91
  #define Z2_MICROSTEPS       16
  #define E0_MAX_CURRENT    1000
  #define E0_SENSE_RESISTOR   91
  #define E0_MICROSTEPS       16
  #define E1_MAX_CURRENT    1000
  #define E1_SENSE_RESISTOR   91
  #define E1_MICROSTEPS       16
  #define E2_MAX_CURRENT    1000
  #define E2_SENSE_RESISTOR   91
  #define E2_MICROSTEPS       16
  #define E3_MAX_CURRENT    1000
  #define E3_SENSE_RESISTOR   91
  #define E3_MICROSTEPS       16
  #define E4_MAX_CURRENT    1000
  #define E4_SENSE_RESISTOR   91
  #define E4_MICROSTEPS       16
#endif

#if ENABLED(HAVE_TMC2130)
  #define R_SENSE           0.11
  #define HOLD_MULTIPLIER    0.5
  #define INTERPOLATE          1
  #define X_CURRENT         1000
  #define X_MICROSTEPS        16
  #define Y_CURRENT         1000
  #define Y_MICROSTEPS        16
  #define Z_CURRENT         1000
  #define Z_MICROSTEPS        16
  #define STEALTHCHOP
#endif
```

```
#if ENABLED(HAVE_L6470DRIVER)
  #define X_MICROSTEPS      16
  #define X_K_VAL           50
  #define X_OVERCURRENT   2000
  #define X_STALLCURRENT  1500
  #define X2_MICROSTEPS     16
  #define X2_K_VAL          50
  #define X2_OVERCURRENT  2000
  #define X2_STALLCURRENT 1500
  #define Y_MICROSTEPS      16
  #define Y_K_VAL           50
  #define Y_OVERCURRENT   2000
  #define Y_STALLCURRENT  1500
  #define Y2_MICROSTEPS     16
  #define Y2_K_VAL          50
  #define Y2_OVERCURRENT  2000
  #define Y2_STALLCURRENT 1500
  #define Z_MICROSTEPS      16
  #define Z_K_VAL           50
  #define Z_OVERCURRENT   2000
  #define Z_STALLCURRENT  1500
  #define Z2_MICROSTEPS     16
  #define Z2_K_VAL          50
  #define Z2_OVERCURRENT  2000
  #define Z2_STALLCURRENT 1500
  #define E0_MICROSTEPS     16
  #define E0_K_VAL          50
  #define E0_OVERCURRENT  2000
  #define E0_STALLCURRENT 1500
  #define E1_MICROSTEPS     16
  #define E1_K_VAL          50
  #define E1_OVERCURRENT  2000
  #define E1_STALLCURRENT 1500
  #define E2_MICROSTEPS     16
  #define E2_K_VAL          50
  #define E2_OVERCURRENT  2000
  #define E2_STALLCURRENT 1500
  #define E3_MICROSTEPS     16
  #define E3_K_VAL          50
  #define E3_OVERCURRENT  2000
  #define E3_STALLCURRENT 1500
  #define E4_MICROSTEPS     16
  #define E4_K_VAL          50
  #define E4_OVERCURRENT  2000
  #define E4_STALLCURRENT 1500
#endif

#define I2C_SLAVE_ADDRESS  0

#define SPINDLE_LASER_ENABLE
#if ENABLED(SPINDLE_LASER_ENABLE)
  #define SPINDLE_LASER_ENABLE_INVERT   false
  #define SPINDLE_LASER_PWM             true
  #define SPINDLE_LASER_PWM_INVERT      false
  #define SPINDLE_LASER_POWERUP_DELAY   5000
  #define SPINDLE_LASER_POWERDOWN_DELAY 5000
  #define SPINDLE_DIR_CHANGE            true
  #define SPINDLE_INVERT_DIR            false
  #define SPINDLE_STOP_ON_DIR_CHANGE    true
  #define SPEED_POWER_SLOPE    118.4
  #define SPEED_POWER_INTERCEPT  0
  #define SPEED_POWER_MIN    500
  #define SPEED_POWER_MAX    30000
#endif

#define EXTENDED_CAPABILITIES_REPORT
#define PROPORTIONAL_FONT_RATIO 1.0
#define FASTER_GCODE_PARSER

#endif
```

8.1.4 dogm_bitmaps.h

```
#define START_BMPHIGH

#if ENABLED(SHOW_BOOTSCREEN)
  #if ENABLED(START_BMPHIGH)
    #define START_BMPWIDTH      112
    #define START_BMPHEIGHT      38
    #define START_BMPBYTEWIDTH   14
    #define START_BMPBYTES      532 // START_BMPWIDTH * START_BMPHEIGHT / 8

    const unsigned char start_bmp[START_BMPBYTES] PROGMEM = {
0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00
,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00
,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00
,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00
,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00
,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00
,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00
,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00
,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00
,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00
,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00
,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00
,0x3f,0xfc,0x3f,0xfe,0x7e,0x0f,0x80,0x00,0xff,0x8f,0xc1,0xf0,0x3f,0xe0
,0x3f,0xfe,0x3f,0xfe,0x7e,0x0f,0x80,0x01,0xff,0xcf,0xc1,0xf0,0x7f,0xf0
,0x3f,0xff,0x3f,0xfe,0x7f,0x0f,0x80,0x03,0xff,0xcf,0xe1,0xf0,0xff,0xf0
,0x3f,0x3f,0x3f,0x00,0x7f,0x8f,0x80,0x07,0xff,0xcf,0xf1,0xf1,0xff,0xf0
,0x3f,0x1f,0xbf,0x00,0x7f,0x8f,0x80,0x0f,0xc1,0x8f,0xf1,0xf3,0xf0,0x60
,0x3f,0x0f,0xbf,0x00,0x7f,0xcf,0x80,0x0f,0xc0,0x0f,0xf9,0xf3,0xf0,0x00
,0x3f,0x0f,0xbf,0x00,0x7f,0xcf,0x80,0x0f,0x80,0x0f,0xf9,0xf3,0xe0,0x00
,0x3f,0x1f,0xbf,0xfc,0x7f,0xef,0x80,0x0f,0x80,0x0f,0xfd,0xf3,0xe0,0x00
,0x3f,0x7f,0x3f,0xfc,0x7f,0xef,0x80,0x0f,0x80,0x0f,0xfd,0xf3,0xe0,0x00
,0x3f,0xff,0x3f,0xfc,0x7d,0xff,0x80,0x0f,0x80,0x0f,0xbf,0xf3,0xe0,0x00
,0x3f,0xfe,0x3f,0x00,0x7d,0xff,0xbf,0xef,0x80,0x0f,0xbf,0xf3,0xe0,0x00
,0x3f,0xf8,0x3f,0x00,0x7c,0xff,0xbf,0xef,0x80,0x0f,0x9f,0xf3,0xe0,0x00
,0x3f,0x00,0x3f,0x00,0x7c,0xff,0xbf,0xcf,0xc0,0x0f,0x9f,0xf3,0xf0,0x00
,0x3f,0x00,0x3f,0x00,0x7c,0x7f,0x80,0x0f,0xc1,0x8f,0x8f,0xf3,0xf0,0x60
,0x3f,0x00,0x3f,0x00,0x7c,0x7f,0x80,0x07,0xff,0xcf,0x8f,0xf1,0xff,0xf0
,0x3f,0x00,0x3f,0xfe,0x7c,0x3f,0x80,0x03,0xff,0xef,0x87,0xf0,0xff,0xf8
,0x3f,0x00,0x3f,0xfe,0x7c,0x3f,0x80,0x01,0xff,0xcf,0x87,0xf0,0x7f,0xf0
,0x3f,0x00,0x3f,0xfe,0x7c,0x1f,0x80,0x00,0xff,0x8f,0x83,0xf0,0x3f,0xe0
,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00
,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00
,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00
,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00
,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00
,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00
,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00
,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00

};
  #else
    #define START_BMPWIDTH      56
    #define START_BMPHEIGHT     19
    #define START_BMPBYTEWIDTH  7
    #define START_BMPBYTES      133 // START_BMPWIDTH * START_BMPHEIGHT / 8

    const unsigned char start_bmp[START_BMPBYTES] PROGMEM = {
      0x1F, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF,
      0x60, 0x00, 0x00, 0x00, 0x00, 0x01, 0xFF,
      0x40, 0x00, 0x00, 0x00, 0x00, 0x00, 0xFF,
      0x80, 0x00, 0x00, 0x00, 0x00, 0x00, 0x7F,
      0x83, 0xCF, 0x00, 0x00, 0x0C, 0x30, 0x3F,
      0x87, 0xFF, 0x80, 0x00, 0x0C, 0x30, 0x1F,
      0x86, 0x79, 0x80, 0x00, 0x0C, 0x00, 0x0F,
      0x8C, 0x30, 0xC7, 0x83, 0x8C, 0x30, 0xE7,
      0x8C, 0x30, 0xCF, 0xC7, 0xCC, 0x31, 0xF3,
      0x8C, 0x30, 0xDC, 0xEC, 0xEC, 0x33, 0xB9,
      0x8C, 0x30, 0xD8, 0x6C, 0x6C, 0x33, 0x19,
      0x8C, 0x30, 0xD0, 0x6C, 0x0C, 0x33, 0x19,
```

```
      0x8C, 0x30, 0xD8, 0x6C, 0x0C, 0x33, 0x19,
      0x8C, 0x30, 0xDC, 0x6C, 0x0E, 0x3B, 0x19,
      0x8C, 0x30, 0xCF, 0x7C, 0x07, 0x9F, 0x19,
      0x8C, 0x30, 0xC7, 0x7C, 0x03, 0x8F, 0x19,
      0x40, 0x00, 0x00, 0x00, 0x00, 0x00, 0x02,
      0x60, 0x00, 0x00, 0x00, 0x00, 0x00, 0x06,
      0x1F, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xF8 };
  #endif
#endif

#if HAS_FAN0
    #define STATUS_SCREENWIDTH      115 //Width in pixels
    #define STATUS_SCREENHEIGHT      19 //Height in pixels
    #define STATUS_SCREENBYTEWIDTH  15 //Width in bytes
    const unsigned char status_screen0_bmp[] PROGMEM = { //AVR-GCC, WinAVR
0x73,0x13,0x0F,//Width pixels,Height pixels,Width bytes
0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0
    x00,0x00,0x00,0x00,0x00,0x00,0x01,0xE0,0x00,0x00,0x00,
0x00,0x00,0x00,0x00,0x00,0x1C,0x40,0xE0,0x00,0x01,0x02,0x10,0x00,0x00,0x00,0x01,
0x00,0x00,0x00,0x00,0x04,0xE0,0x60,0x00,0x00,0x84,0xFE,0x00,0x00,0x00,0x02,0x80,
0x00,0x00,0x00,0x08,0x40,0xAA,0x00,0x00,0x78,0x8E,0x00,0x00,0x00,0x04,0x40,0x00,
0x00,0x00,0x1C,0x41,0x0A,0x00,0x00,0x00,0x9A,0x00,0x00,0x00,0x04,0x42,0x00,0x00,
0x00,0x00,0x42,0x04,0x00,0x00,0x00,0xB2,0x00,0x00,0x00,0x04,0x45,0x00,0x00,0x00,
0x00,0x44,0x04,0x00,0x00,0x00,0xE6,0x00,0x00,0x00,0x04,0x48,0x80,0x00,0x00,0x00,
0x48,0x00,0x00,0x00,0x00,0xCE,0x00,0x00,0x00,0x03,0x88,0x80,0x00,0x00,0x02,0x50,
0x00,0x00,0x00,0x00,0x9A,0x00,0x00,0x00,0x00,0x08,0x80,0x00,0x00,0x01,0x60,0x00,
0x00,0x00,0x00,0xB2,0x00,0x00,0x00,0x10,0x08,0x80,0x00,0x00,0x00,0xC0,0x00,0x00,
0x00,0x00,0xE6,0x00,0x00,0x00,0x28,0x27,0x00,0x00,0x00,0x00,0xE0,0x00,0x00,0x00,
0x00,0xCE,0x00,0x00,0x00,0x44,0x50,0x00,0x00,0x00,0x01,0x50,0x00,0x00,0x00,0x00,
0xFE,0x00,0x00,0x00,0x44,0x88,0x00,0x00,0x00,0x02,0x48,0x14,0x00,0x00,0x00,0x44,
0x00,0x00,0x00,0x44,0x88,0x00,0x00,0x00,0x00,0x44,0x14,0x00,0x00,0x00,0x28,0x00,
0x00,0x00,0x44,0x88,0x00,0x00,0x00,0x00,0x02,0x88,0x00,0x00,0x00,0x10,0x00,0x00,
0x00,0x38,0x88,0x00,0x00,0x00,0x00,0x01,0x94,0x00,0x00,0x00,0x10,0x00,0x00,0x00,
0x00,0x70,0x00,0x00,0x00,0x00,0x03,0x94,0x00,0x00,0x00,0x10,0x00,0x00,0x00,0x00,
0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,

    };

    #define STATUS_SCREENWIDTH      115 //Width in pixels
    #define STATUS_SCREENHEIGHT      19 //Height in pixels
    #define STATUS_SCREENBYTEWIDTH  15 //Width in bytes
    const unsigned char status_screen1_bmp[] PROGMEM = { //AVR-GCC, WinAVR
0x73,0x14,0x0F,//Width pixels,Height pixels,Width bytes
0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0
    x00,0x00,0x00,0x00,0x00,0x00,0x01,0xE0,0x00,0x00,0x00,
//0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00
    ,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
0x00,0x00,0x00,0x00,0x00,0x1C,0x40,0xE0,0x00,0x01,0x02,0x10,0x00,0x00,0x00,0x01,
0x00,0x00,0x00,0x00,0x04,0xE0,0x60,0x00,0x00,0x84,0xFE,0x00,0x00,0x00,0x02,0x80,
0x00,0x00,0x00,0x08,0x40,0xAA,0x00,0x00,0x78,0xA6,0x00,0x00,0x00,0x04,0x40,0x00,
0x00,0x00,0x1C,0x41,0x0A,0x00,0x00,0x00,0xCE,0x00,0x00,0x00,0x04,0x42,0x00,0x00,
0x00,0x00,0x42,0x04,0x00,0x00,0x00,0x9A,0x00,0x00,0x00,0x04,0x45,0x00,0x00,0x00,
0x00,0x44,0x04,0x00,0x00,0x00,0xB2,0x00,0x00,0x00,0x04,0x48,0x80,0x00,0x00,0x00,
0x48,0x00,0x00,0x00,0x00,0xE6,0x00,0x00,0x00,0x03,0x88,0x80,0x00,0x00,0x02,0x50,
0x00,0x00,0x00,0x00,0xCE,0x00,0x00,0x00,0x00,0x08,0x80,0x00,0x00,0x01,0x60,0x00,
0x00,0x00,0x00,0x9A,0x00,0x00,0x00,0x10,0x08,0x80,0x00,0x00,0x00,0xC0,0x00,0x00,
0x00,0x00,0xB2,0x00,0x00,0x00,0x28,0x27,0x00,0x00,0x00,0x00,0xE0,0x00,0x00,0x00,
0x00,0xE2,0x00,0x00,0x00,0x44,0x50,0x00,0x00,0x00,0x01,0x50,0x00,0x00,0x00,0x00,
0xFE,0x00,0x00,0x00,0x44,0x88,0x00,0x00,0x00,0x02,0x48,0x14,0x00,0x00,0x00,0x44,
0x00,0x00,0x00,0x44,0x88,0x00,0x00,0x00,0x00,0x44,0x14,0x00,0x00,0x00,0x28,0x00,
0x00,0x00,0x44,0x88,0x00,0x00,0x00,0x00,0x02,0x88,0x00,0x00,0x00,0x91,0x00,0x00,
0x00,0x38,0x88,0x00,0x00,0x00,0x00,0x01,0x94,0x00,0x00,0x00,0x52,0x00,0x00,0x00,
0x00,0x70,0x00,0x00,0x00,0x00,0x03,0x94,0x00,0x00,0x01,0x14,0x80,0x00,0x00,0x00,
0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0xC3,0x00,0x00,

    };
    #define STATUS_SCREENWIDTH      115 //Width in pixels
    #define STATUS_SCREENHEIGHT      19 //Height in pixels
    #define STATUS_SCREENBYTEWIDTH  15 //Width in bytes
    const unsigned char status_screen2_bmp[] PROGMEM = { //AVR-GCC, WinAVR
 0x73,0x13,0x0F,//Width pixels,Height pixels,Width bytes
```

```
0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0
    x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x01,
0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x02,0x80,
0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x04,0x40,0x00,
0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x04,0x42,0x00,0x00,
0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x04,0x45,0x00,0x00,0x00,
0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x04,0x48,0x80,0x00,0x00,0x00,
0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x03,0x88,0x80,0x00,0x00,0x00,0x00,
0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x08,0x80,0x00,0x00,0x00,0x00,0x00,
0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x10,0x08,0x80,0x00,0x00,0x00,0x00,0x00,0x00,
0x00,0x00,0x00,0x00,0x00,0x00,0x28,0x27,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
0x00,0x00,0x00,0x00,0x00,0x44,0x50,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
0x00,0x00,0x00,0x00,0x44,0x88,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
0x00,0x00,0x00,0x44,0x88,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
0x00,0x00,0x44,0x88,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
0x00,0x38,0x88,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
0x00,0x70,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,

    };
    #define STATUS_SCREENWIDTH      115 //Width in pixels
    #define STATUS_SCREENHEIGHT      19 //Height in pixels
    #define STATUS_SCREENBYTEWIDTH  15 //Width in bytes
    const unsigned char status_screen3_bmp[] PROGMEM = { //AVR-GCC, WinAVR
 0x73,0x13,0x0F,//Width pixels,Height pixels,Width bytes
0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0
    x00,0x00,0x00,0x00,0x00,0x00,0x00,0x01,0xE0,0x00,0x00,0x00,
0x00,0x00,0x00,0x00,0x00,0x1C,0x40,0xE0,0x00,0x01,0x02,0x10,0x00,0x00,0x00,0x01,
0x00,0x00,0x00,0x00,0x04,0xE0,0x60,0x00,0x00,0x84,0xFE,0x00,0x00,0x00,0x02,0x80,
0x00,0x00,0x00,0x08,0x40,0xAA,0x00,0x00,0x78,0x8E,0x00,0x00,0x00,0x04,0x40,0x00,
0x00,0x00,0x1C,0x41,0x0A,0x00,0x00,0x00,0x9A,0x00,0x00,0x00,0x04,0x42,0x00,0x00,
0x00,0x00,0x42,0x04,0x00,0x00,0x00,0xB2,0x00,0x00,0x00,0x04,0x45,0x00,0x00,0x00,
0x00,0x44,0x04,0x00,0x00,0x00,0xE6,0x00,0x00,0x00,0x04,0x48,0x80,0x00,0x00,0x00,
0x48,0x00,0x00,0x00,0x00,0xCE,0x00,0x00,0x00,0x03,0x88,0x80,0x00,0x00,0x02,0x50,
0x00,0x00,0x00,0x00,0x9A,0x00,0x00,0x00,0x00,0x08,0x80,0x00,0x00,0x01,0x60,0x00,
0x00,0x00,0x00,0xB2,0x00,0x00,0x00,0x10,0x08,0x80,0x00,0x00,0x00,0xC0,0x00,0x00,
0x00,0x00,0xE6,0x00,0x00,0x00,0x28,0x27,0x00,0x00,0x00,0x00,0xE0,0x00,0x00,0x00,
0x00,0xCE,0x00,0x00,0x00,0x44,0x50,0x00,0x00,0x00,0x01,0x50,0x00,0x00,0x00,0x00,
0xFE,0x00,0x00,0x00,0x44,0x88,0x00,0x00,0x00,0x02,0x48,0x14,0x00,0x00,0x00,0x44,
0x00,0x00,0x00,0x44,0x88,0x00,0x00,0x00,0x00,0x44,0x14,0x00,0x00,0x00,0x28,0x00,
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0x00,0x70,0x00,0x00,0x00,0x00,0x03,0x94,0x00,0x00,0x00,0x10,0x00,0x00,0x00,0x00,
0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,

    };

#endif
```
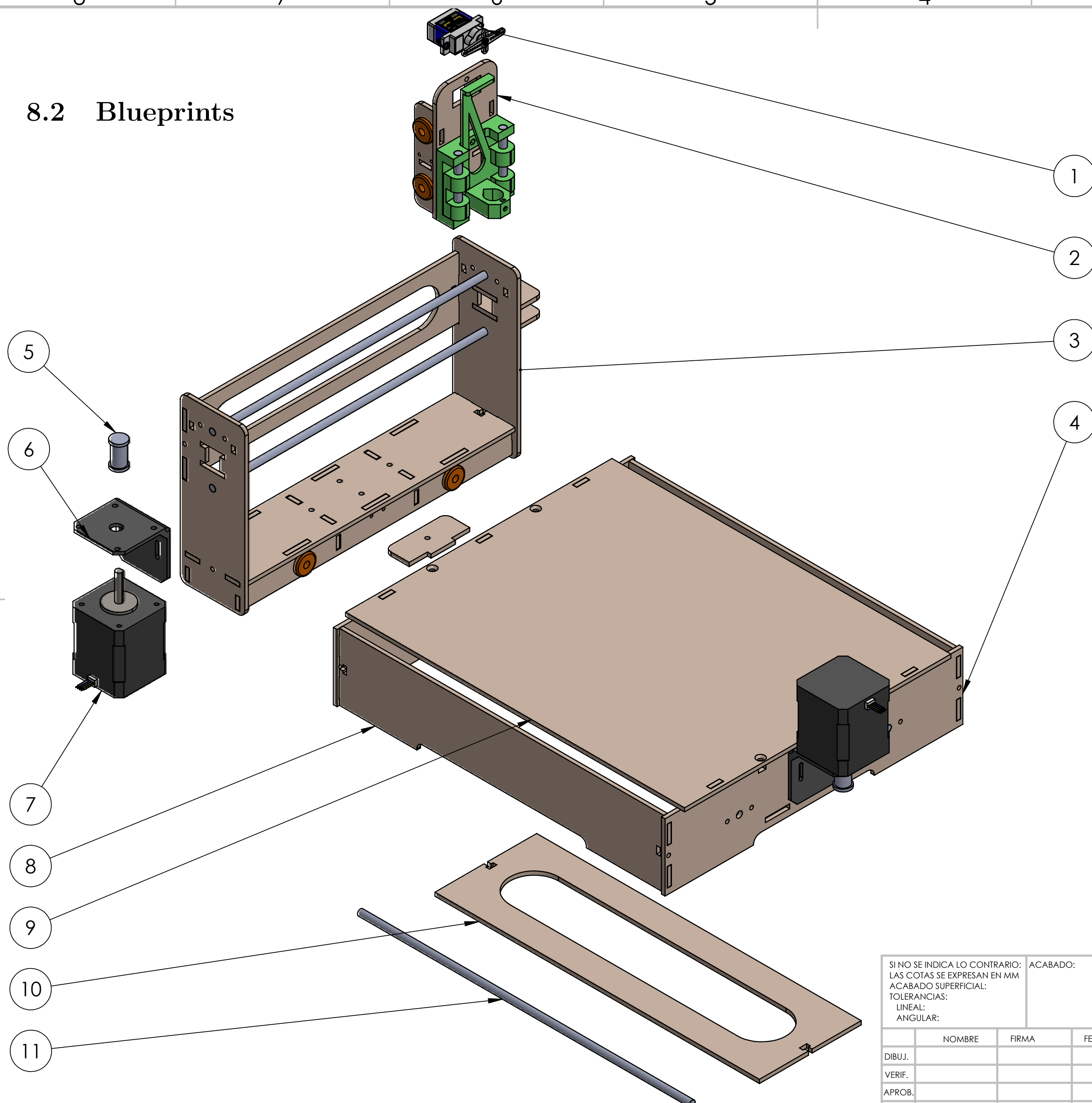

8.2 Blueprints



N.º DE ELEMENTO	N.º DE PIEZA	CANTIDAD
1	SG90 - Micro Servo 9g - Tower Pro.STEP	1
2	EJE Z	1
3	EJE Y	1
4	Frontales_CNC	2
5	Buje_Motor_CNC	2
6	Base_Motor_CNC	2
7	Nema 17 Stepper Motor 17HD48002H-22B.step	2
8	Laterales_CNC	2
9	Base_CNC	1
10	Soporte_CNC	1
11	CilindroX_CNC	2
12	BASERODAMIENTO_CNC	2

SI NO SE INDICA LO CONTRARIO:
LAS COTAS SE EXPRESAN EN MM
ACABADO SUPERFICIAL:
TOLERANCIAS:
LINEAL:
ANGULAR:

ACABADO:

REBARBAR Y ROMPER ARISTAS VIVAS

NO CAMBIE LA ESCALA

REVISIÓN

	NOMBRE	FIRMA	FECHA
DIBUJ.			
VERIF.			
APROB.			
FABR.			
CALID.			

TÍTULO:

MATERIAL:

N.º DE DIBUJO

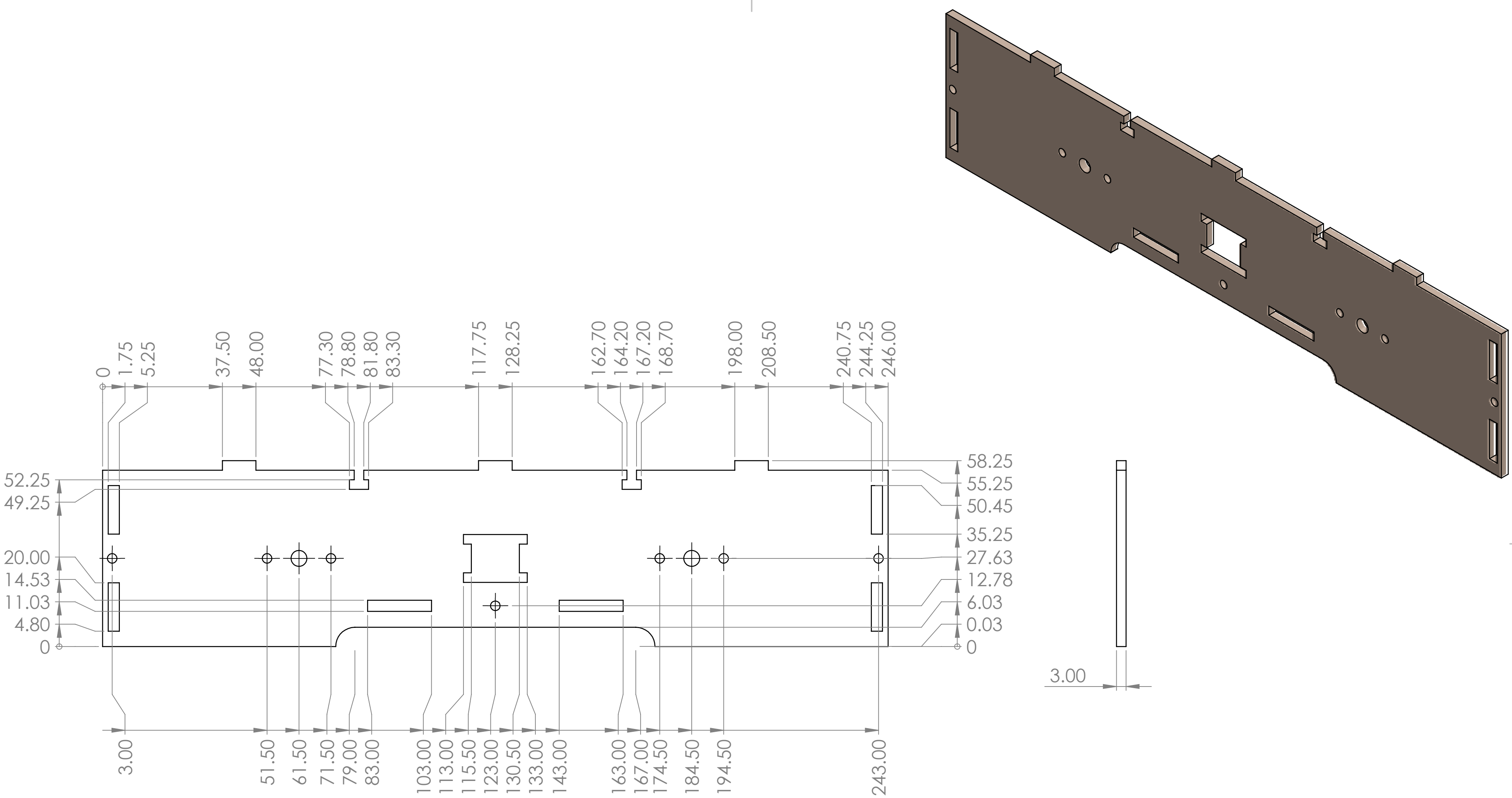
CNC_FINAL

A3

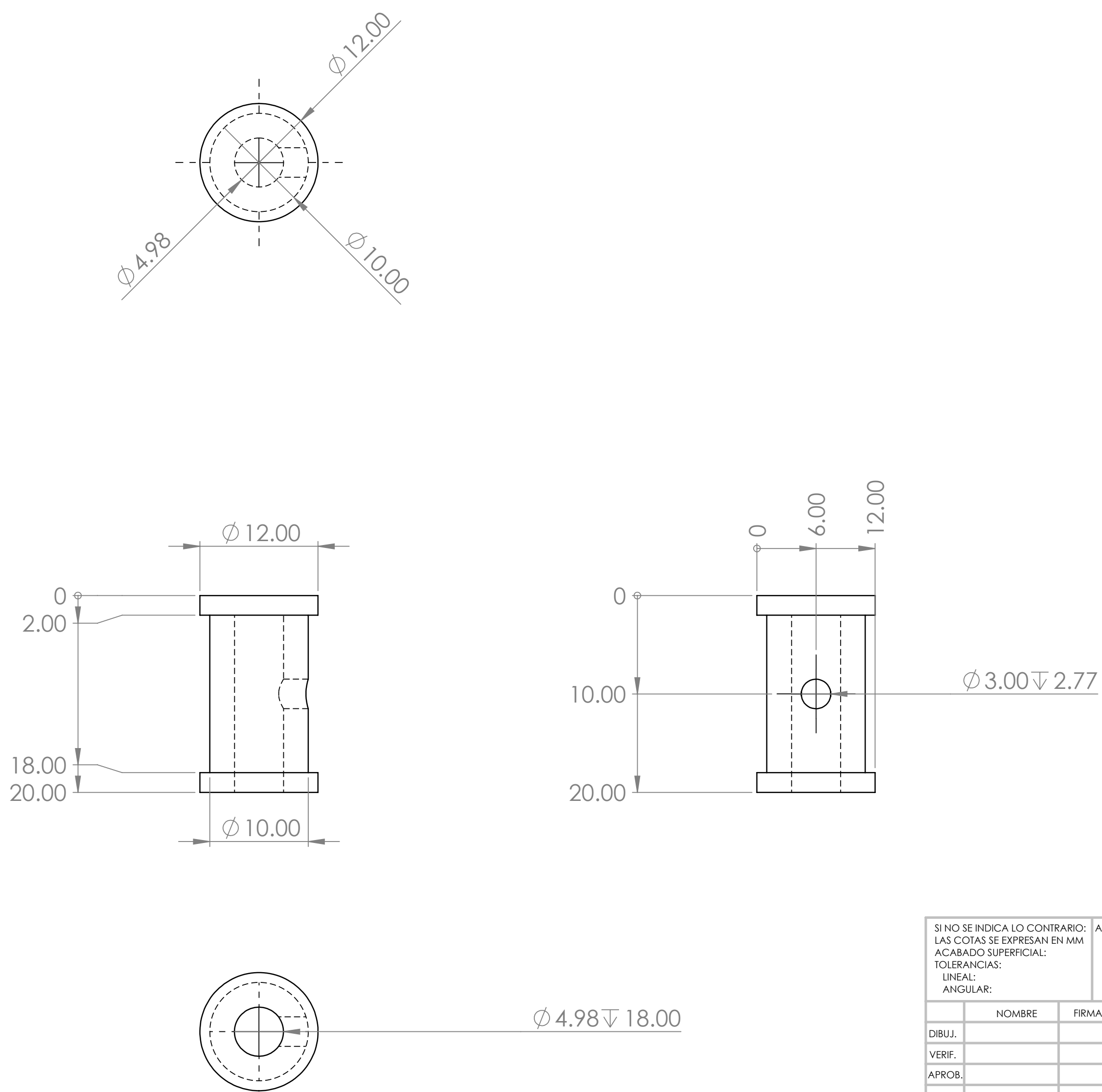
PESO:

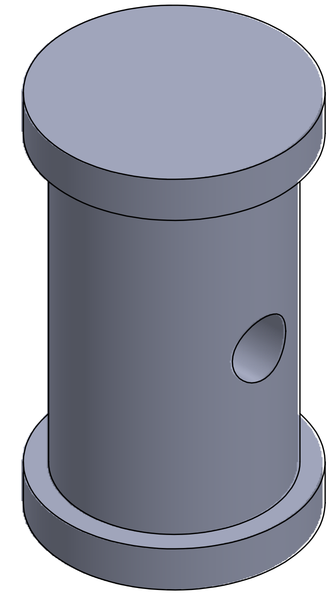
ESCALA:1:10

HOJA 1 DE 1

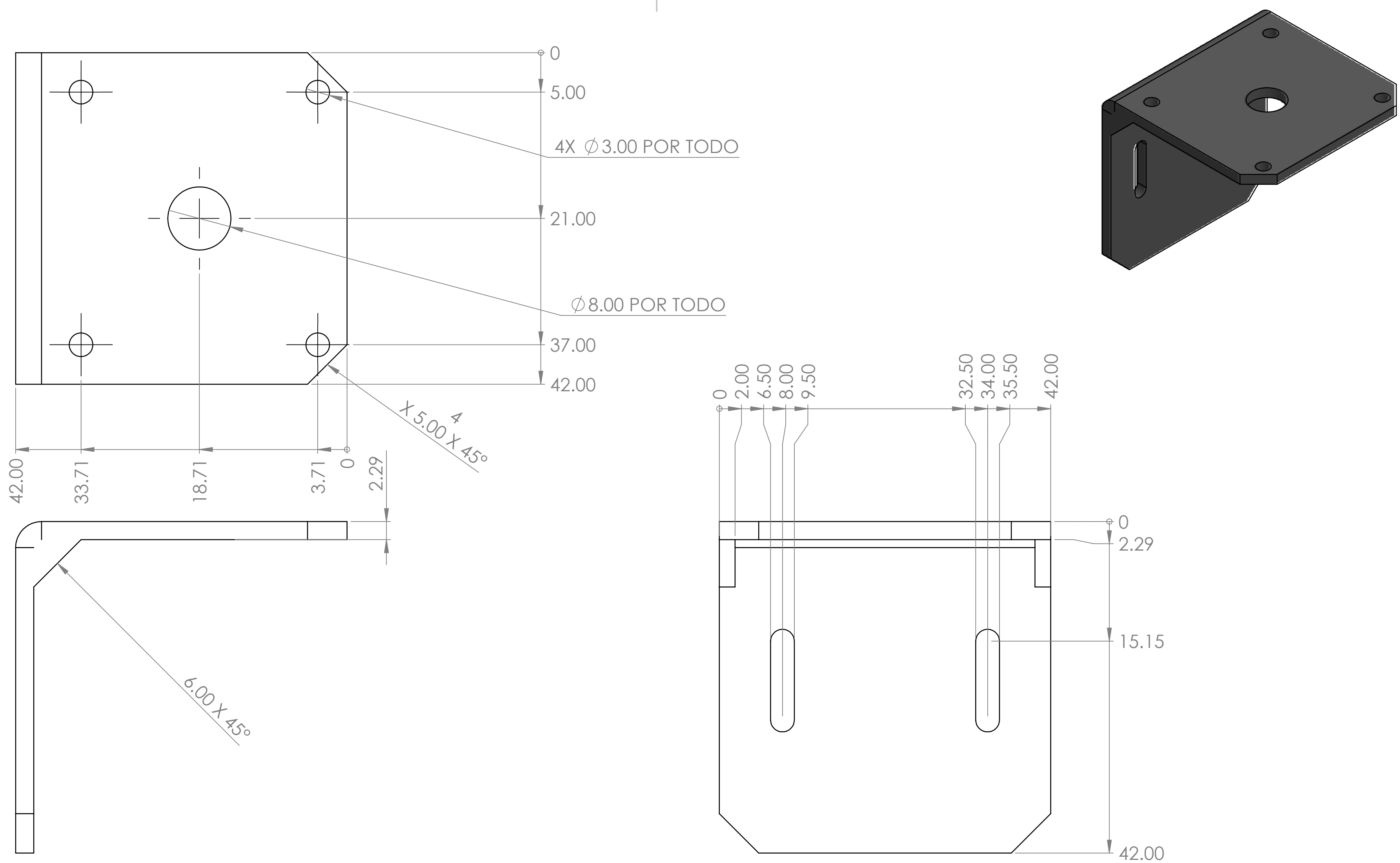


SI NO SE INDICA LO CONTRARIO: LAS COTAS SE EXPRESAN EN MM ACABADO SUPERFICIAL: TOLERANCIAS: LINEAL: ANGULAR:	ACABADO:	REBARBAR Y ROMPER ARISTAS VIVAS	NO CAMBIE LA ESCALA	REVISIÓN	
	NOMBRE	FIRMA	FECHA		TÍTULO:
DIBUJ.					
VERIF.					
APROB.					
FABR.					
CALID.				MATERIAL:	N.º DE DIBUJO Frontales_CNC
					A3
				PESO:	ESCALA:1:2
					HOJA 1 DE 1





SI NO SE INDICA LO CONTRARIO: LAS COTAS SE EXPRESAN EN MM ACABADO SUPERFICIAL: TOLERANCIAS: LINEAL: ANGULAR:	ACABADO:		REBARBAR Y ROMPER ARISTAS VIVAS	NO CAMBIE LA ESCALA	REVISIÓN
	NOMBRE	FIRMA	FECHA	TÍTULO:	
DIBUJ.					
VERIF.					
APROB.					
FABR.					
CALID.			MATERIAL:	N.º DE DIBUJO Buje_Motor_CNC	A3
			PESO:	ESCALA:5:1	HOJA 1 DE 1



SI NO SE INDICA LO CONTRARIO:
LAS COTAS SE EXPRESAN EN MM
ACABADO SUPERFICIAL:
TOLERANCIAS:
LINEAL:
ANGULAR:

ACABADO:

REBARBAR Y ROMPER ARISTAS VIVAS

NO CAMBIE LA ESCALA

REVISIÓN

	NOMBRE	FIRMA	FECHA
DIBUJ.			
VERIF.			
APROB.			
FABR.			
CALID.			

TÍTULO:

MATERIAL:

N.º DE DIBUJO

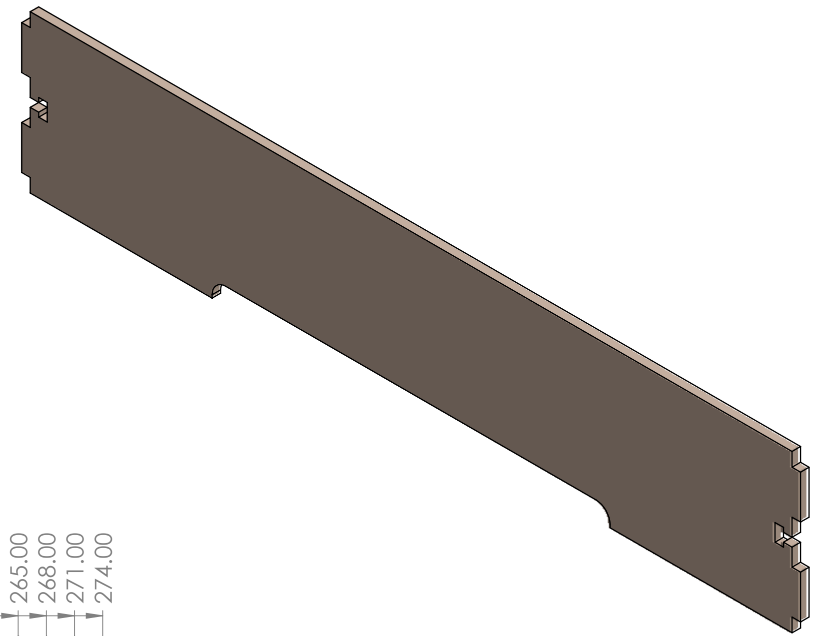
Base_Motor_CNC

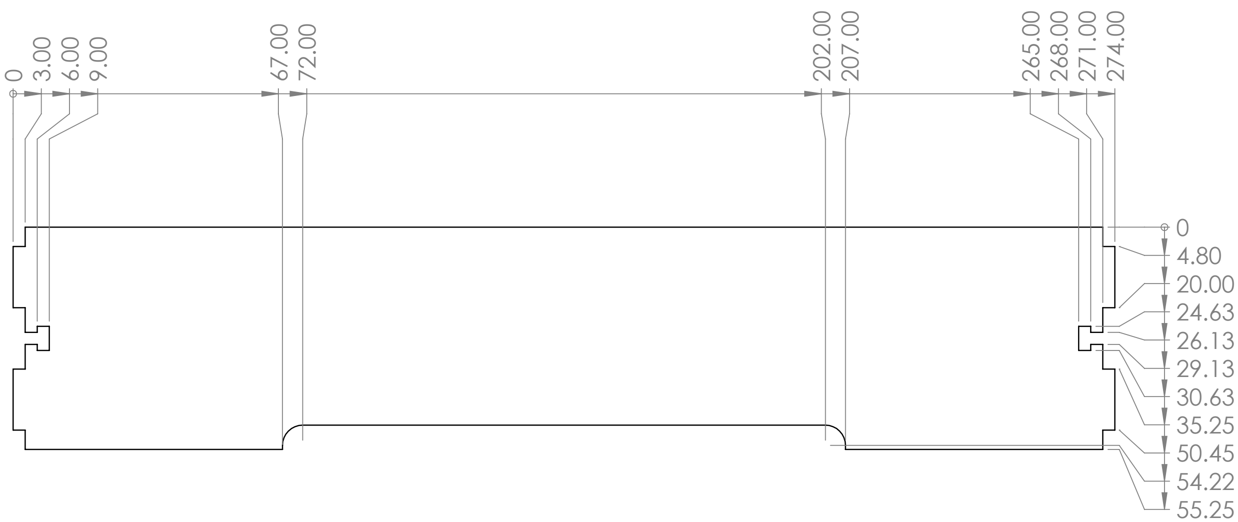
A3

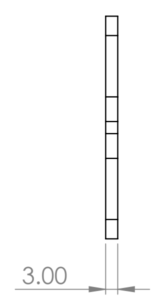
PESO:

ESCALA:1:1

HOJA 1 DE 1







SI NO SE INDICA LO CONTRARIO:
LAS COTAS SE EXPRESAN EN MM
ACABADO SUPERFICIAL:
TOLERANCIAS:
LINEAL:
ANGULAR:

ACABADO:

REBARBAR Y ROMPER ARISTAS VIVAS

NO CAMBIE LA ESCALA

REVISIÓN

	NOMBRE	FIRMA	FECHA		
DIBUJ.					
VERIF.					
APROB.					
FABR.					
CALID.					

TÍTULO:

MATERIAL:

N.º DE DIBUJO

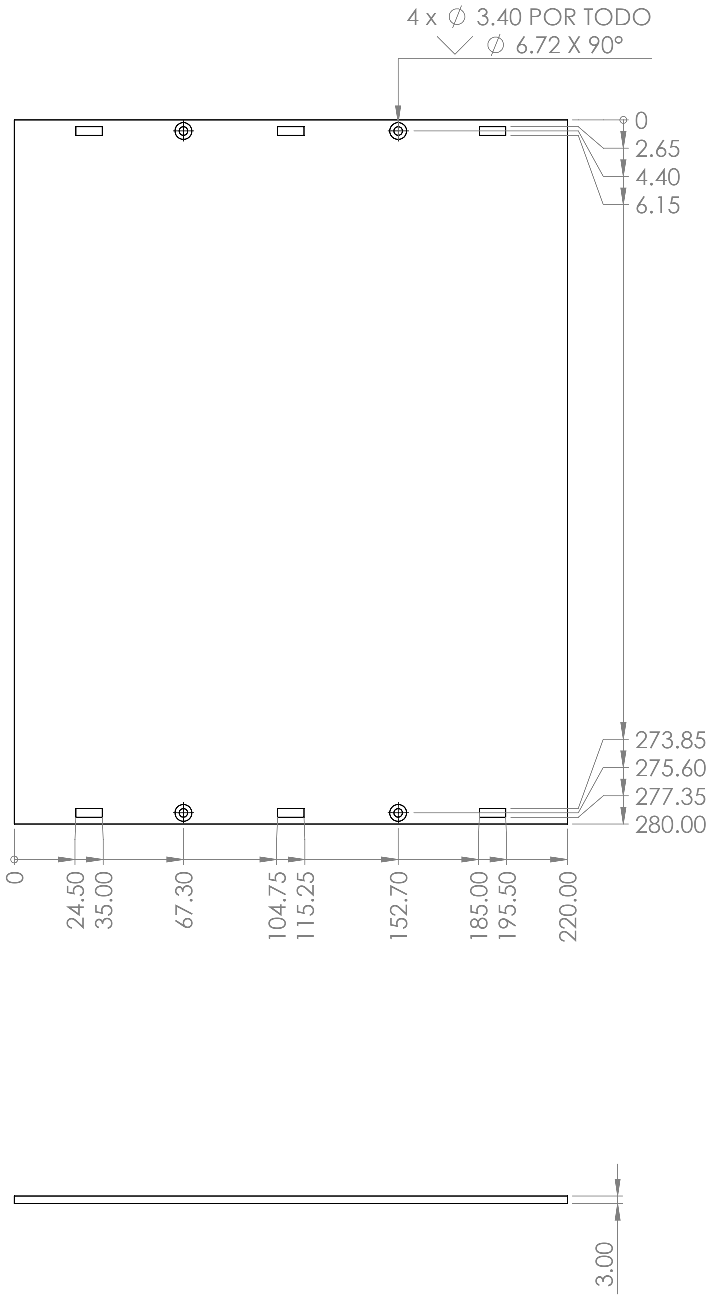
Laterales_CNC

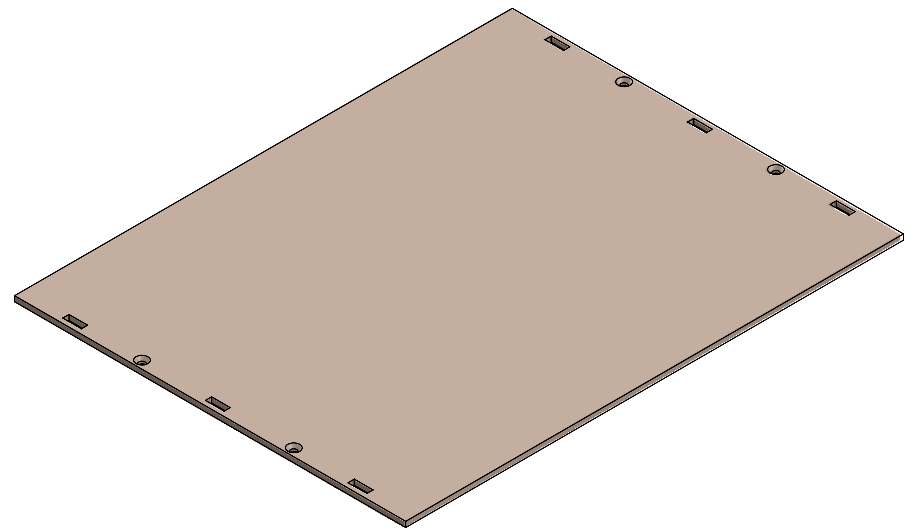
A3

PESO:

ESCALA:1:2

HOJA 1 DE 1





SI NO SE INDICA LO CONTRARIO:
LAS COTAS SE EXPRESAN EN MM
ACABADO SUPERFICIAL:
TOLERANCIAS:
LINEAL:
ANGULAR:

ACABADO:

REBARBAR Y ROMPER ARISTAS VIVAS

NO CAMBIE LA ESCALA

REVISIÓN

	NOMBRE	FIRMA	FECHA
DIBUJ.			
VERIF.			
APROB.			
FABR.			
CALID.			

MATERIAL:

PESO:

TÍTULO:

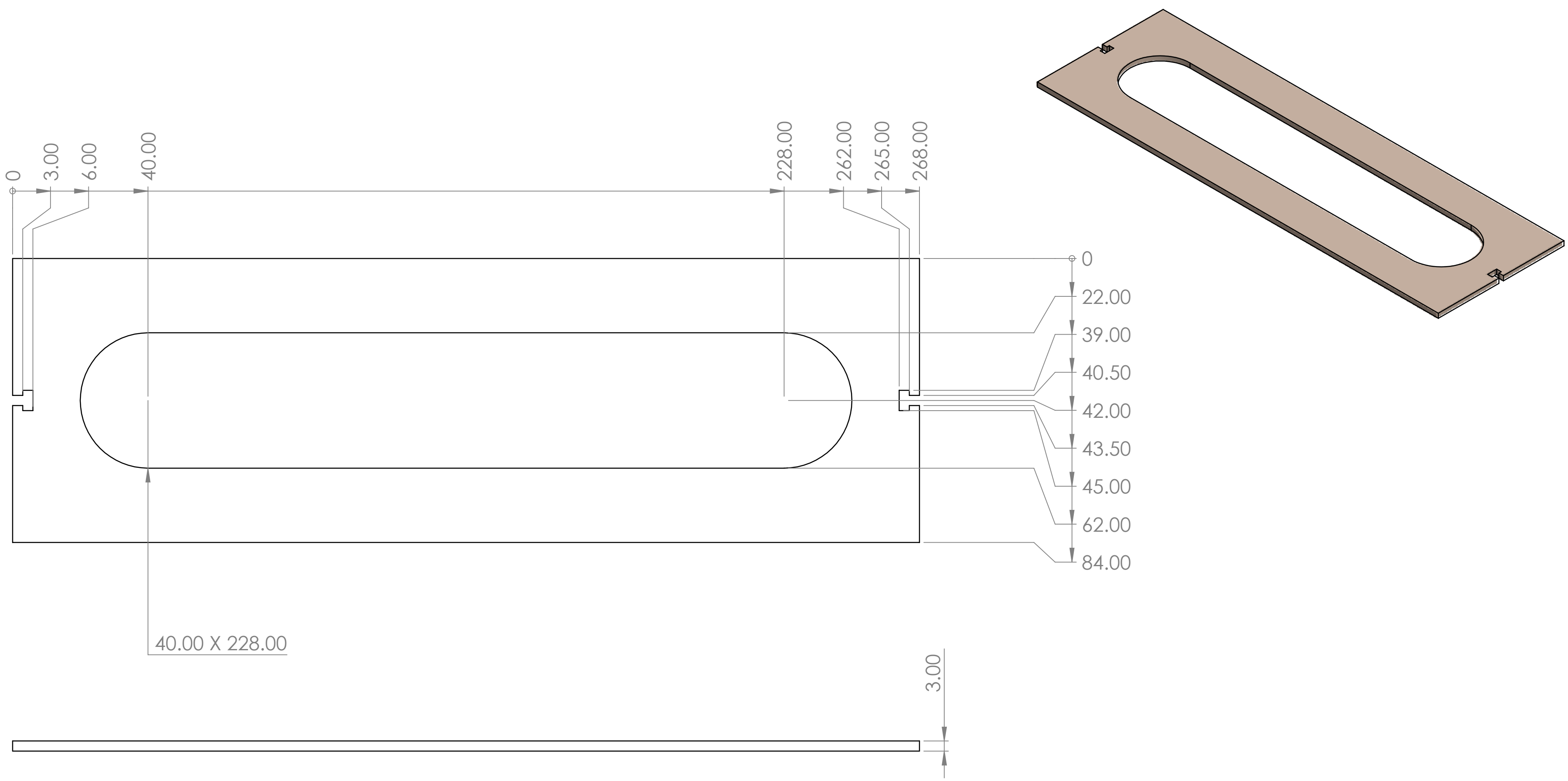
N.º DE DIBUJO

Base_CNC

A3

ESCALA:1:5

HOJA 1 DE 1

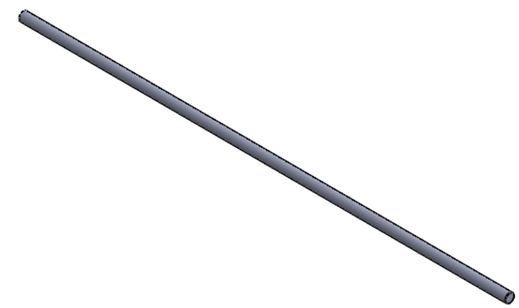


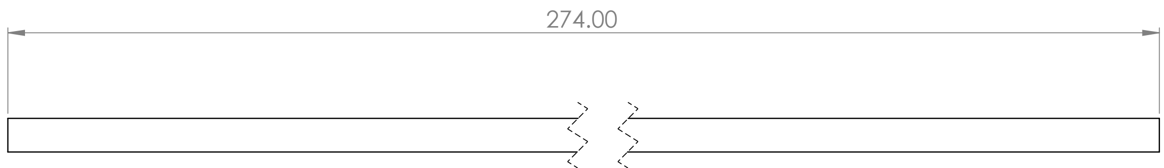
SI NO SE INDICA LO CONTRARIO: LAS COTAS SE EXPRESAN EN MM ACABADO SUPERFICIAL: TOLERANCIAS: LINEAL: ANGULAR:	ACABADO:	REBARBAR Y ROMPER ARISTAS VIVAS	NO CAMBIE LA ESCALA	REVISIÓN

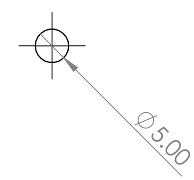
	NOMBRE	FIRMA	FECHA			TÍTULO:
DIBUJ.						
VERIF.						
APROB.						
FABR.						
CALID.				MATERIAL:		N.º DE DIBUJO Soporte_CNC
				PESO:		ESCALA:1:2

A3

HOJA 1 DE 1







SI NO SE INDICA LO CONTRARIO:
LAS COTAS SE EXPRESAN EN MM
ACABADO SUPERFICIAL:
TOLERANCIAS:
LINEAL:
ANGULAR:

ACABADO:

REBARBAR Y ROMPER ARISTAS VIVAS

NO CAMBIE LA ESCALA

REVISIÓN

	NOMBRE	FIRMA	FECHA
DIBUJ.			
VERIF.			
APROB.			
FABR.			
CALID.			

TÍTULO:

MATERIAL:

N.º DE DIBUJO

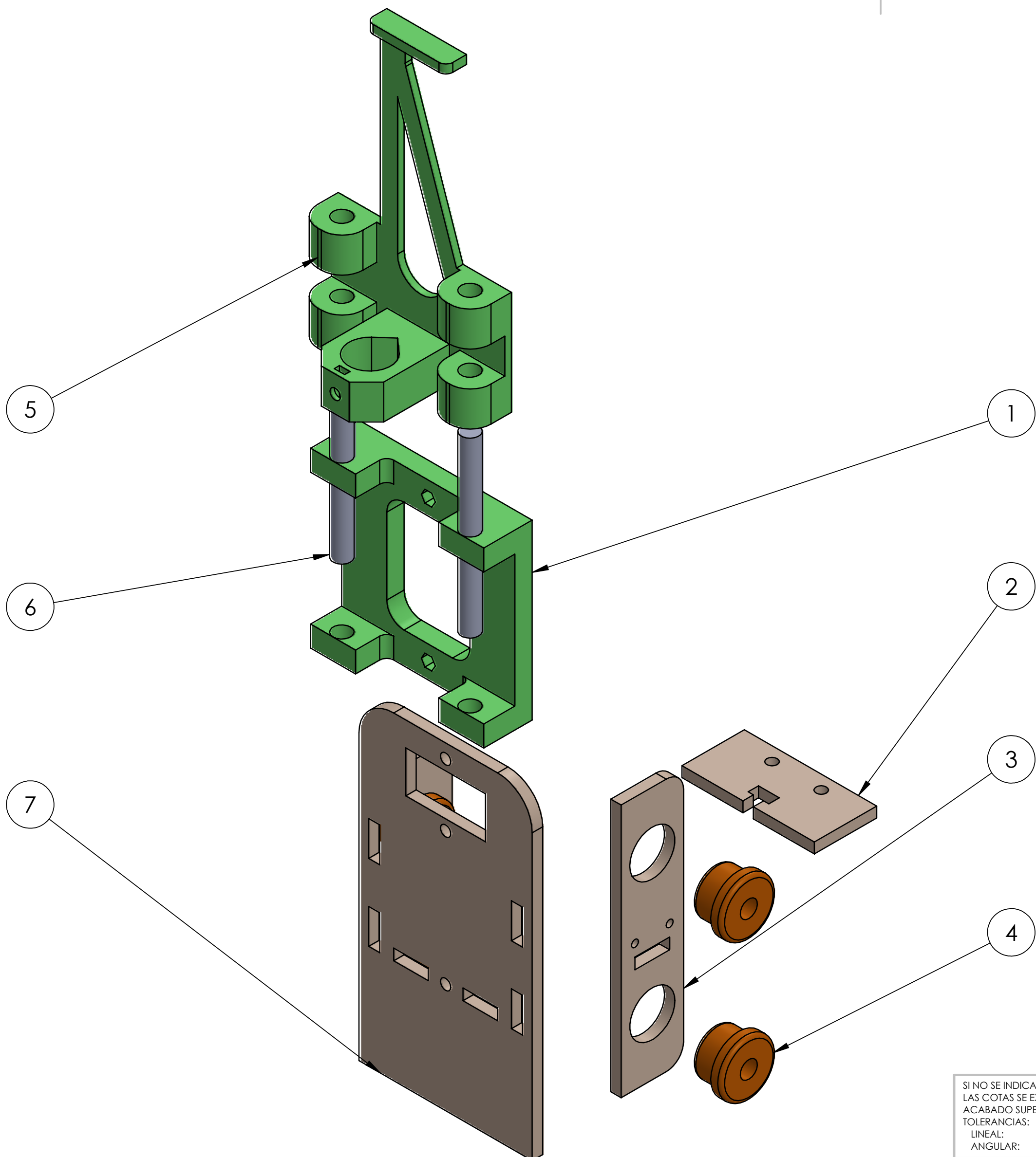
CilindroX_CNC

A3

PESO:

ESCALA:1:2

HOJA 1 DE 1



N.º DE ELEMENTO	N.º DE PIEZA	CANTIDAD
1	BaseLapiz_CNC	1
2	SoporteZP_CNC	1
3	SoporteTZ_CNC	2
4	Bujepr_CNC	4
5	EjeMovil_CNC	1
6	CilindroY_CNC	2
7	SoporteZ_CNC	1

SI NO SE INDICA LO CONTRARIO:
LAS COTAS SE EXPRESAN EN MM
ACABADO SUPERFICIAL:
TOLERANCIAS:
LINEAL:
ANGULAR:

ACABADO:

REBARBAR Y ROMPER ARISTAS VIVAS

NO CAMBIE LA ESCALA

REVISIÓN

	NOMBRE	FIRMA	FECHA
DIBUJ.			
VERIF.			
APROB.			
FABR.			
CALID.			

MATERIAL:

PESO:

TÍTULO:

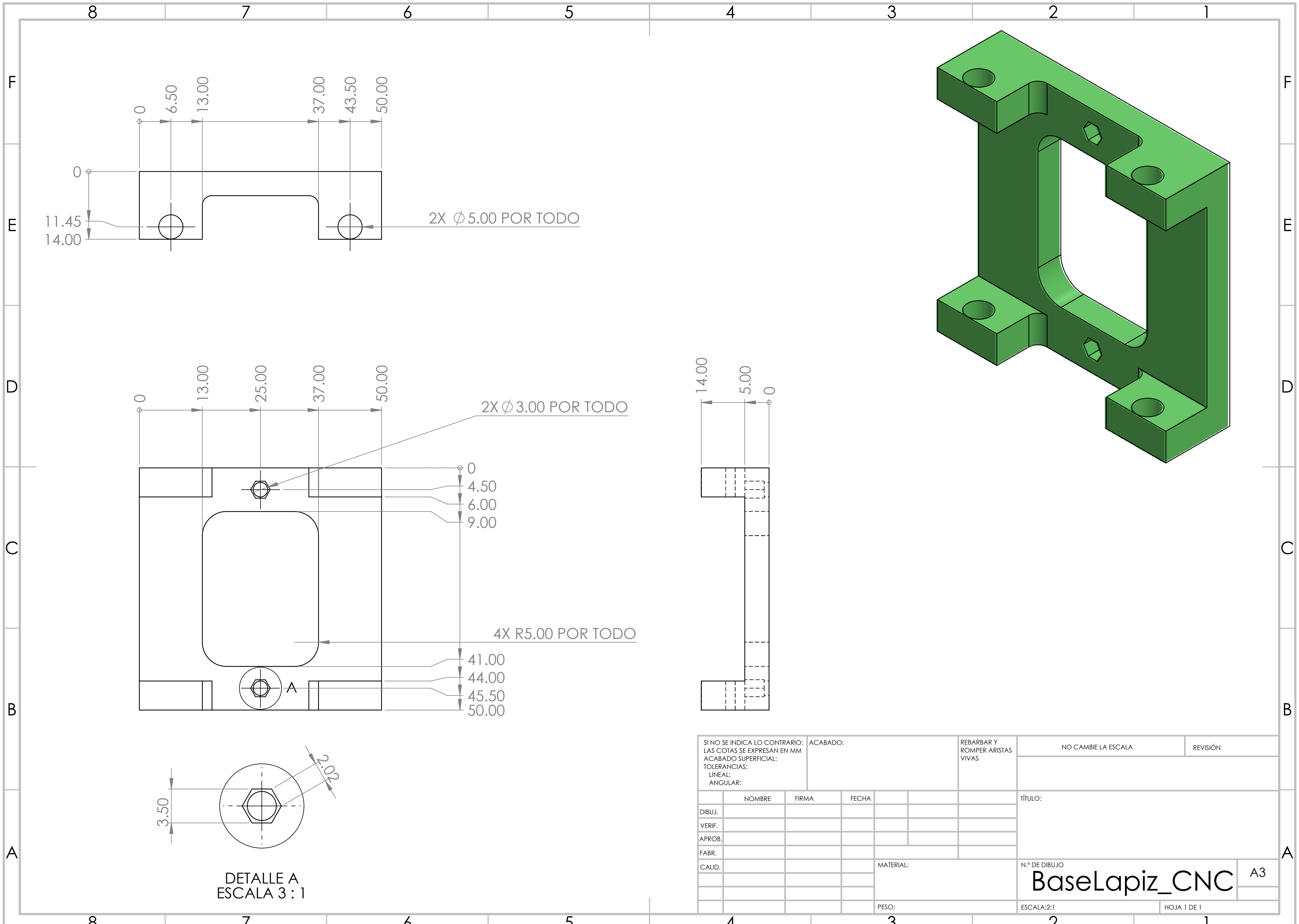
N.º DE DIBUJO

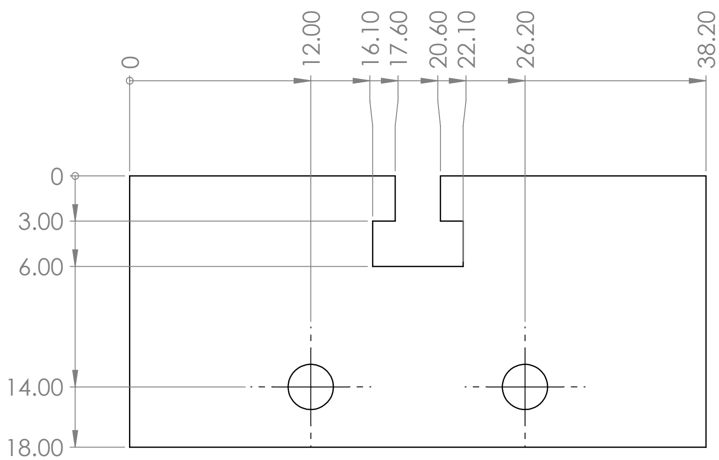
EJE Z

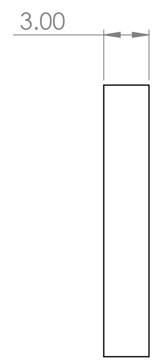
A3

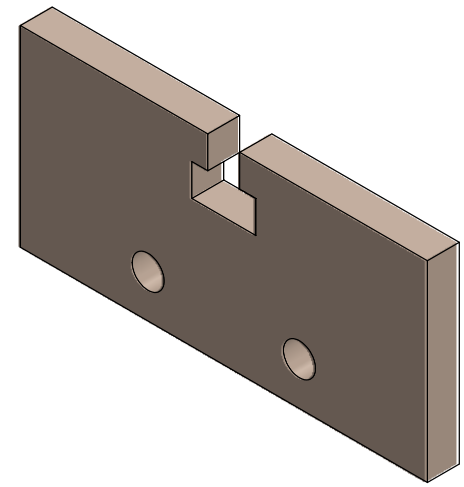
ESCALA:1:5

HOJA 1 DE 1

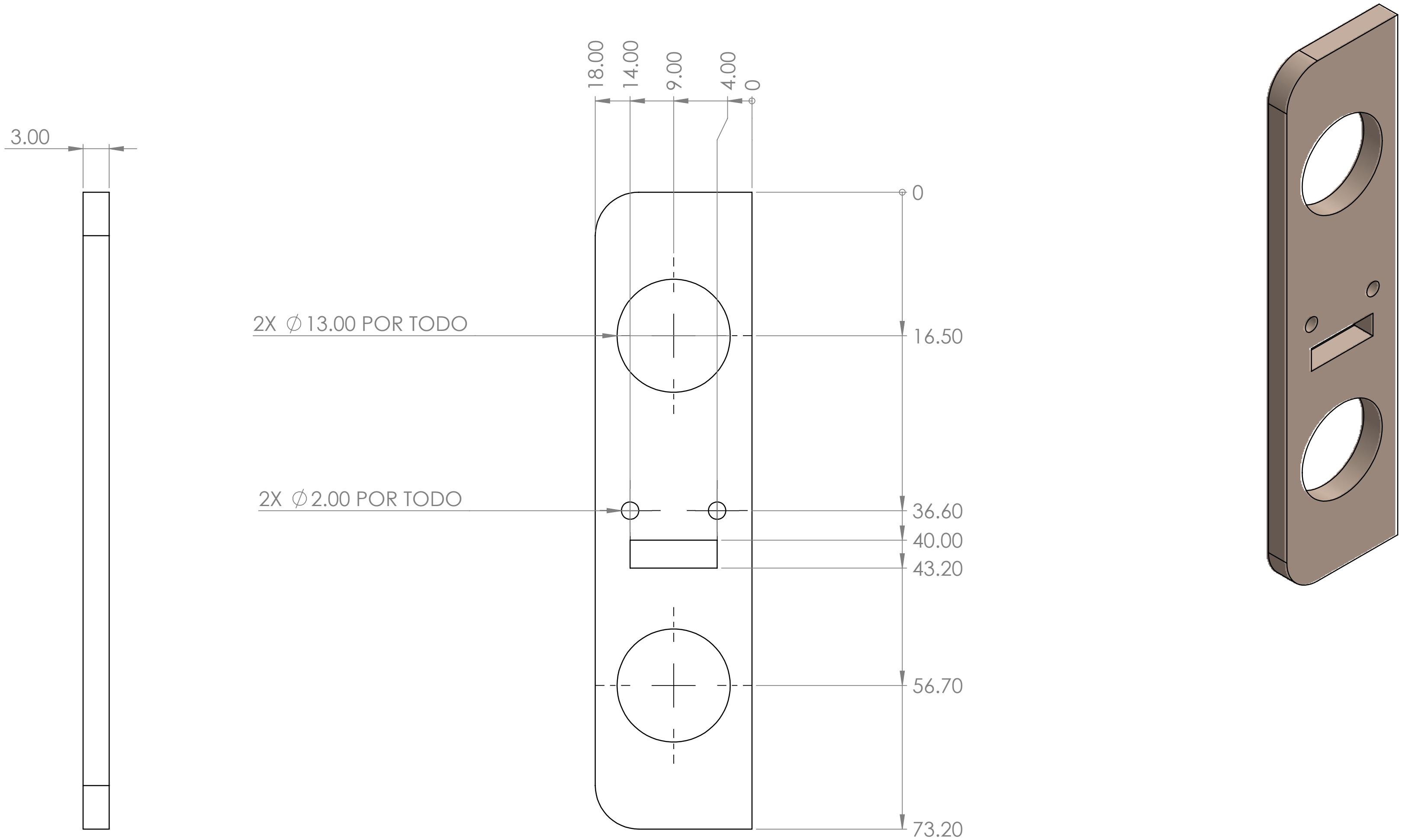

0
6.50
13.00
37.00
43.50
50.00
0
11.45
14.00
2X ∅5.00 POR TODO
0
13.00
25.00
37.00
50.00
2X ∅3.00 POR TODO
0
4.50
6.00
9.00
4X R5.00 POR TODO
41.00
44.00
45.50
50.00
A
14.00
5.00
0
3.50
2.02
DETALLE A
ESCALA 3 : 1
SI NO SE INDICA LO CONTRARIO:
LAS COTAS SE EXPRESAN EN MM
ACABADO SUPERFICIAL:
TOLERANCIAS:
LINEAL:
ANGULAR:
ACABADO:
REBARBAR Y ROMPER ARISTAS VIVAS
NO CAMBIE LA ESCALA
REVISIÓN
NOMBRE
FIRMA
FECHA
DIBUJ.
VERIF.
APROB.
FABR.
CALID.
TÍTULO:
MATERIAL:
N.º DE DIBUJO
BaseLapiz_CNC
A3
PESO:
ESCALA:2:1
HOJA 1 DE 1





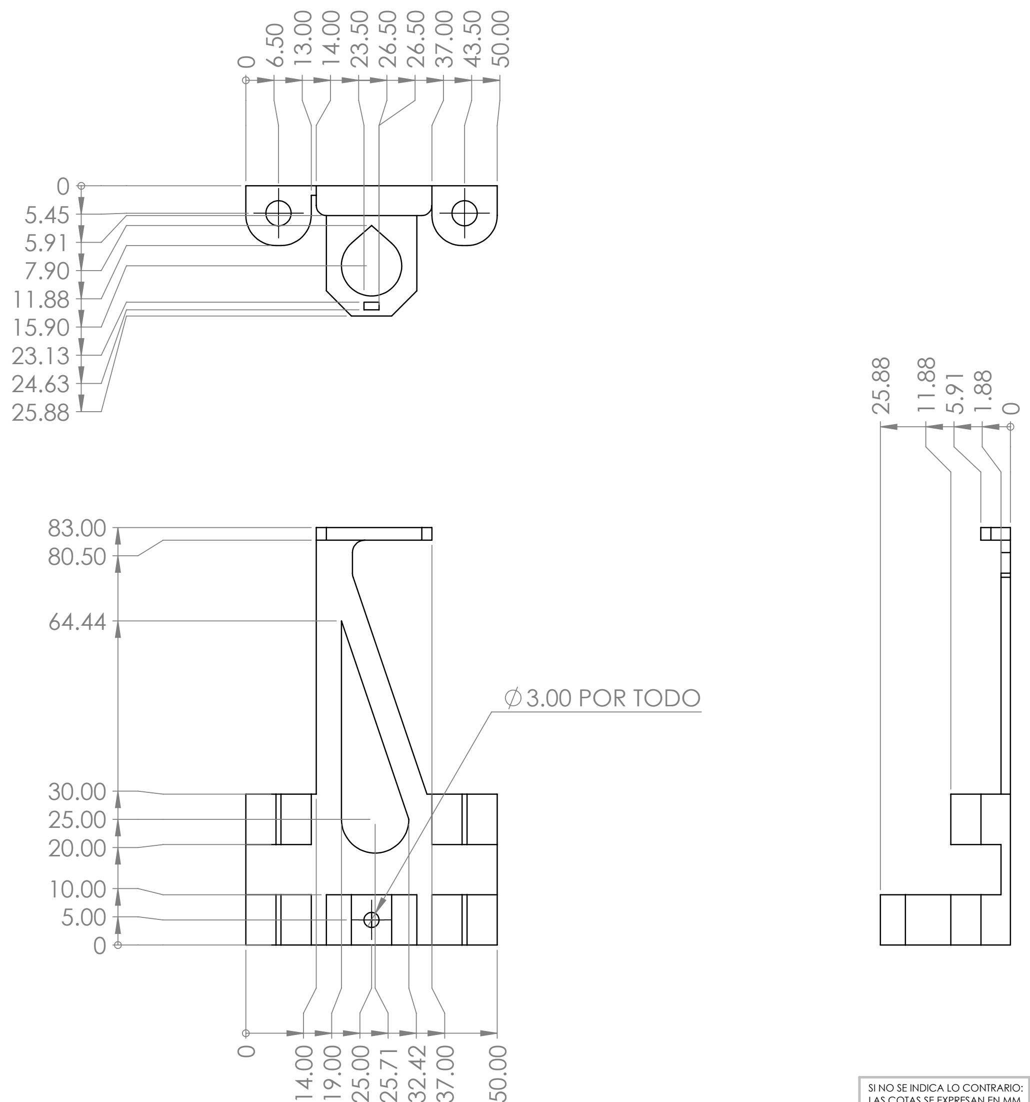


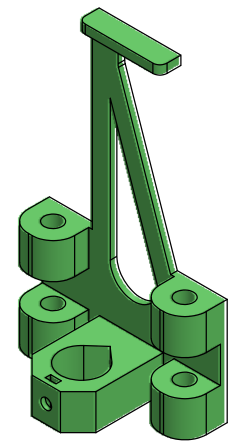
SI NO SE INDICA LO CONTRARIO: LAS COTAS SE EXPRESAN EN MM ACABADO SUPERFICIAL: TOLERANCIAS: LINEAL: ANGULAR:	ACABADO:		REBARBAR Y ROMPER ARISTAS VIVAS	NO CAMBIE LA ESCALA	REVISIÓN	
	NOMBRE	FIRMA	FECHA	TÍTULO:		
DIBUJ.						
VERIF.						
APROB.						
FABR.						
CALID.			MATERIAL:	N.º DE DIBUJO SoporteZP_CNC	A3	
			PESO:	ESCALA:5:1	HOJA 1 DE 1	



SI NO SE INDICA LO CONTRARIO: LAS COTAS SE EXPRESAN EN MM ACABADO SUPERFICIAL: TOLERANCIAS: LINEAL: ANGULAR:	ACABADO:	REBARBAR Y ROMPER ARISTAS VIVAS	NO CAMBIE LA ESCALA	REVISIÓN

	NOMBRE	FIRMA	FECHA	
DIBUJ.				TÍTULO:
VERIF.				
APROB.				
FABR.				
CALID.			MATERIAL:	N.º DE DIBUJO SoporteTZ_CNC A3
			PESO:	ESCALA:1:1 HOJA 1 DE 1





SI NO SE INDICA LO CONTRARIO:
LAS COTAS SE EXPRESAN EN MM
ACABADO SUPERFICIAL:
TOLERANCIAS:
LINEAL:
ANGULAR:

ACABADO:

REBARBAR Y ROMPER ARISTAS VIVAS

NO CAMBIE LA ESCALA

REVISIÓN

	NOMBRE	FIRMA	FECHA
DIBUJ.			
VERIF.			
APROB.			
FABR.			
CALID.			

TÍTULO:

MATERIAL:

N.º DE DIBUJO

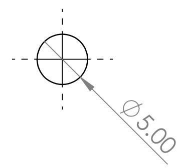
EjeMovil_CNC

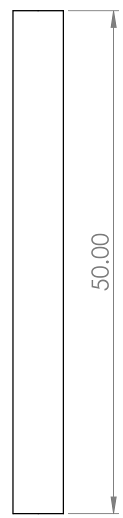
A3

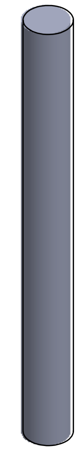
PESO:

ESCALA:1:1

HOJA 1 DE 1







SI NO SE INDICA LO CONTRARIO:
LAS COTAS SE EXPRESAN EN MM
ACABADO SUPERFICIAL:
TOLERANCIAS:
LINEAL:
ANGULAR:

ACABADO:

REBARBAR Y ROMPER ARISTAS VIVAS

NO CAMBIE LA ESCALA

REVISIÓN

	NOMBRE	FIRMA	FECHA
DIBUJ.			
VERIF.			
APROB.			
FABR.			
CALID.			

TÍTULO:

MATERIAL:

N.º DE DIBUJO

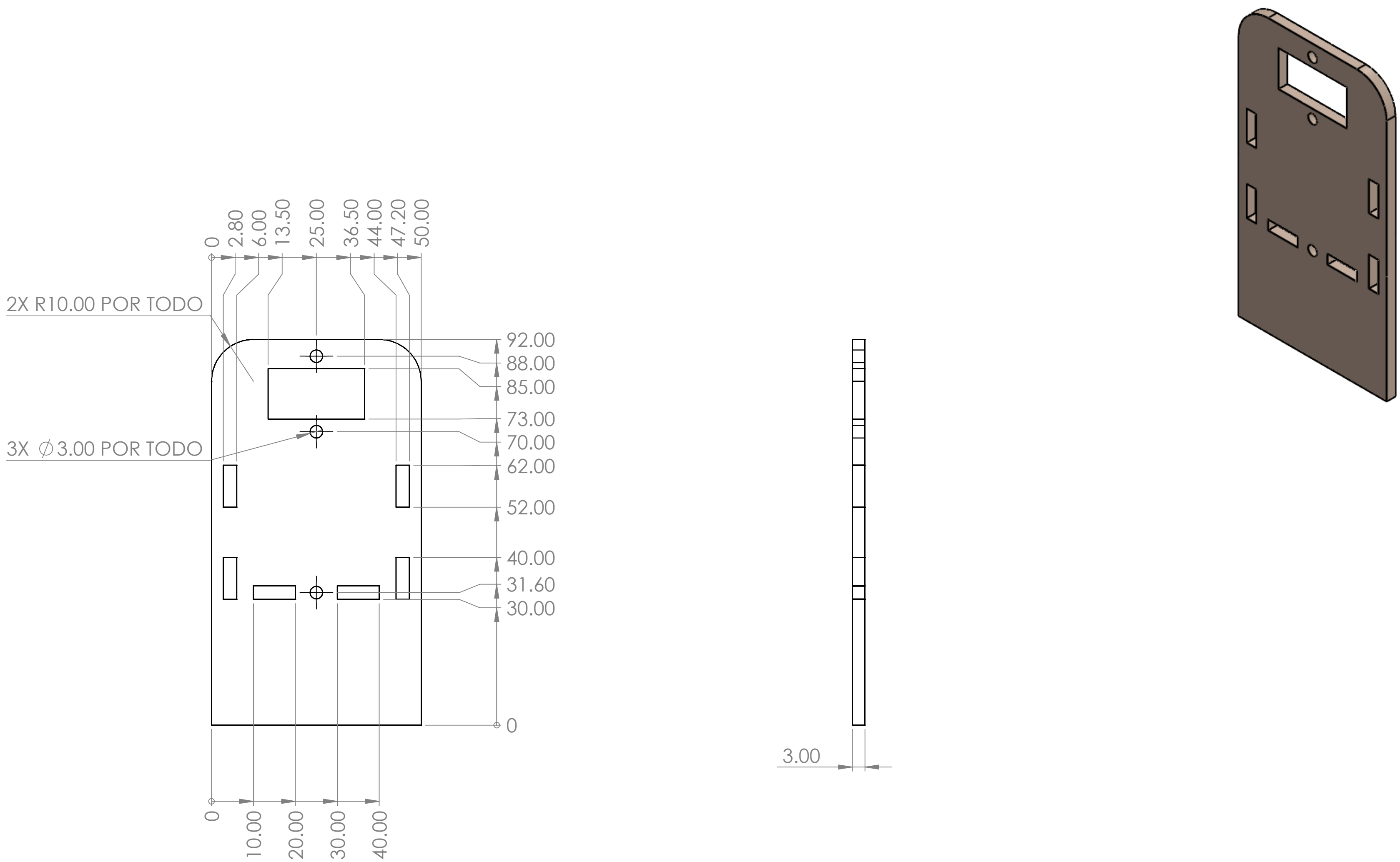
CilindroY_CNC

A3

PESO:

ESCALA:2:1

HOJA 1 DE 1



SI NO SE INDICA LO CONTRARIO: LAS COTAS SE EXPRESAN EN MM ACABADO SUPERFICIAL: TOLERANCIAS: LINEAL: ANGULAR:	ACABADO:	REBARBAR Y ROMPER ARISTAS VIVAS	NO CAMBIE LA ESCALA
			REVISIÓN

	NOMBRE	FIRMA	FECHA
DIBUJ.			
VERIF.			
APROB.			
FABR.			
CALID.			

TÍTULO:

MATERIAL:

PESO:

N.º DE DIBUJO

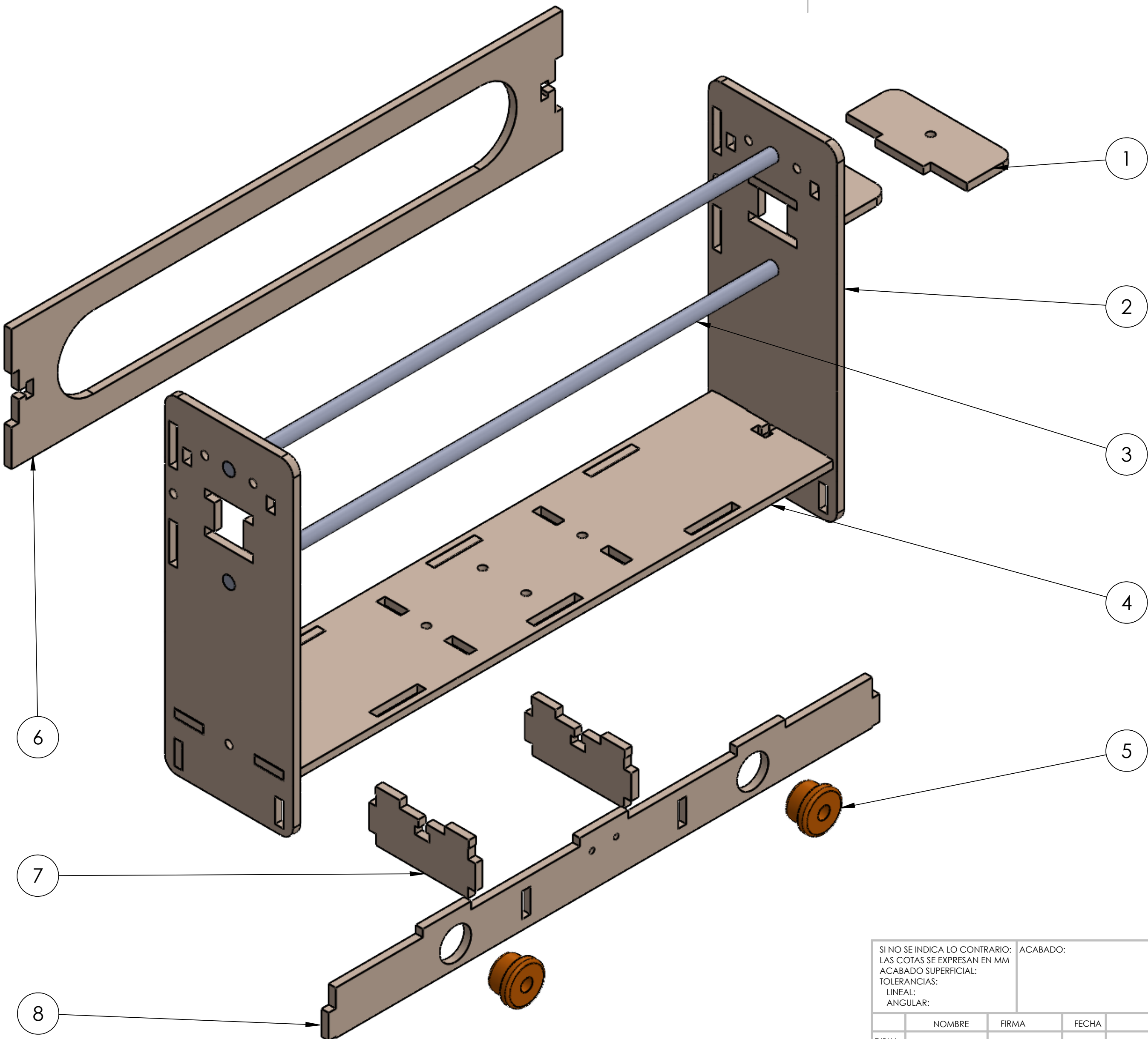
SoporteZ_CNC

A3

ESCALA:1:1

HOJA 1 DE 1

N.º DE ELEMENTO	N.º DE PIEZA	CANTIDAD
1	BASERODAMIENTO_CNC	2
2	SoporteY_CNC	2
3	EjeCY_CNC	2
4	BaseY	1
5	Bujepr_CNC	4
6	FronteY_CNC	1
7	SoportePequeno_CNC	2
8	SoporteBuje_CNC	2



SI NO SE INDICA LO CONTRARIO: LAS COTAS SE EXPRESAN EN MM ACABADO SUPERFICIAL: TOLERANCIAS: LINEAL: ANGULAR:

ACABADO:

REBARBAR Y ROMPER ARISTAS VIVAS

NO CAMBIE LA ESCALA

REVISIÓN

	NOMBRE	FIRMA	FECHA
DIBUJ.			
VERIF.			
APROB.			
FABR.			
CALID.			

TÍTULO:

MATERIAL:

N.º DE DIBUJO

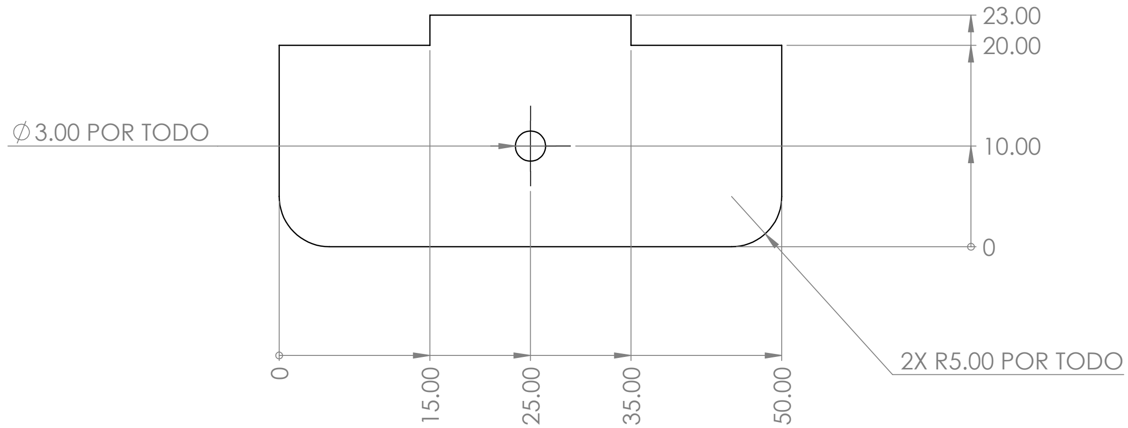
EJE Y

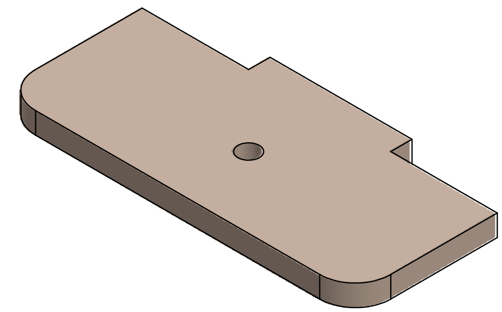
A3

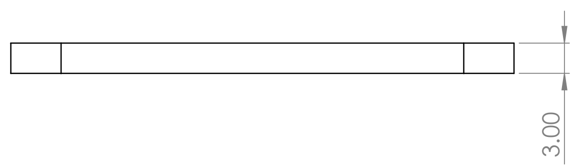
PESO:

ESCALA:1:5

HOJA 1 DE 1







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LAS COTAS SE EXPRESAN EN MM
ACABADO SUPERFICIAL:
TOLERANCIAS:
LINEAL:
ANGULAR:

ACABADO:

REBARBAR Y ROMPER ARISTAS VIVAS

NO CAMBIE LA ESCALA

REVISIÓN

	NOMBRE	FIRMA	FECHA
DIBUJ.			
VERIF.			
APROB.			
FABR.			
CALID.			

TÍTULO:

MATERIAL:

N.º DE DIBUJO

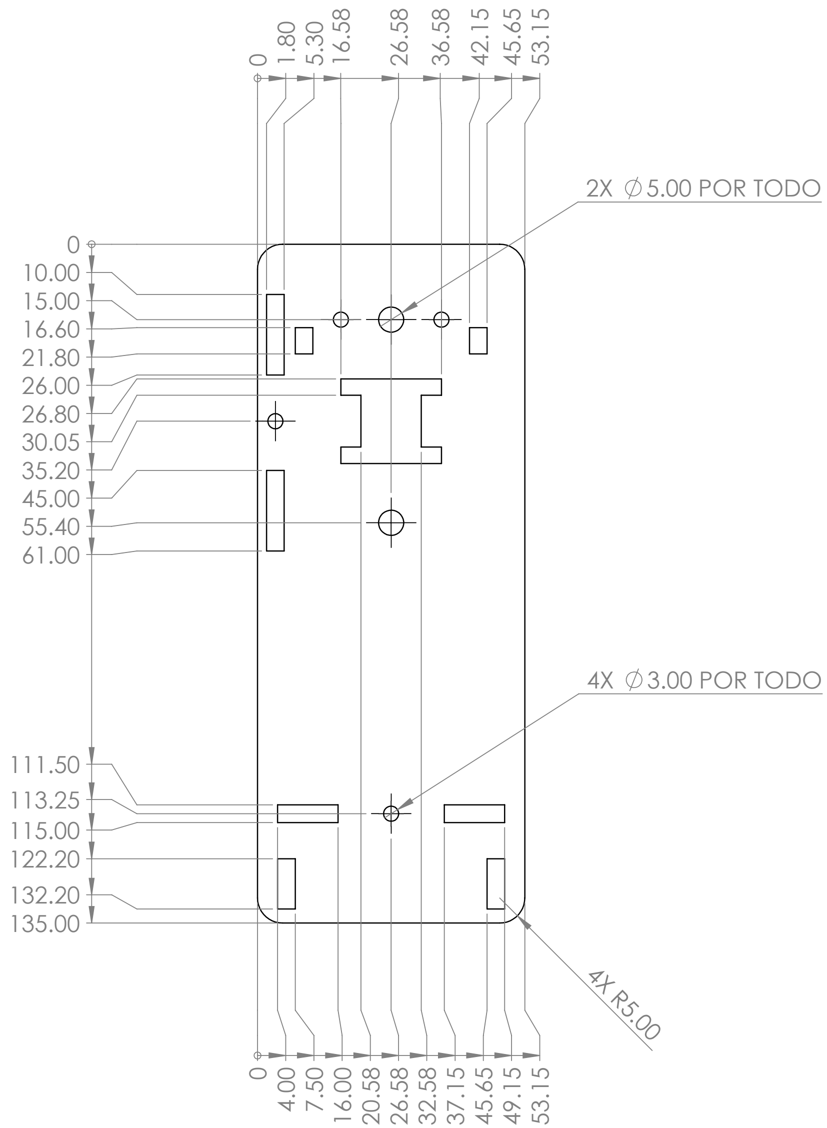
BASERODAMIENTO_CNC

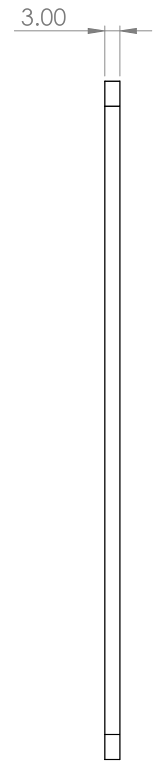
A3

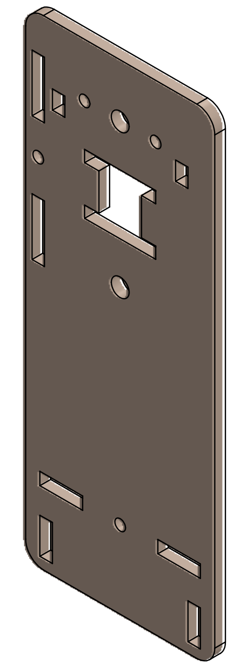
PESO:

ESCALA:2:1

HOJA 1 DE 1







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DIBUJ.						
VERIF.						
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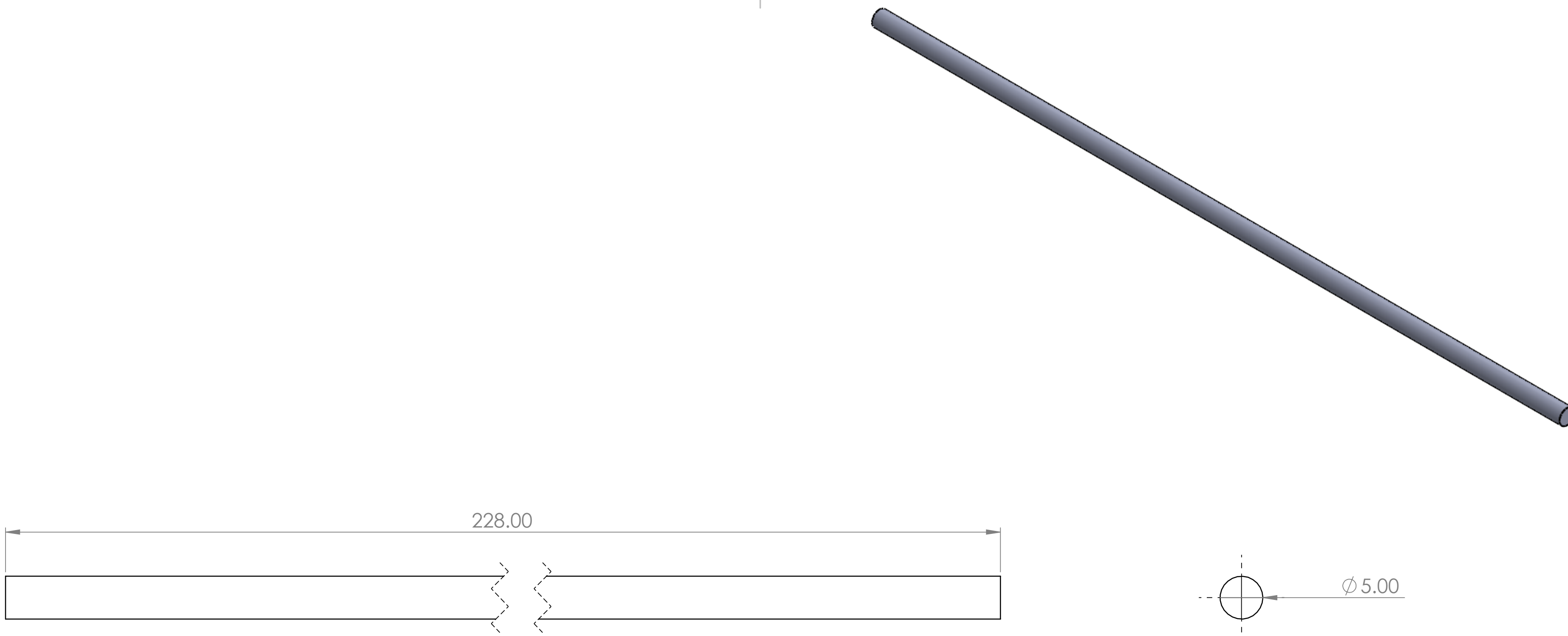
HOJA 1 DE 1



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LINEAL:
ANGULAR:

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REBARBAR Y ROMPER ARISTAS VIVAS

NO CAMBIE LA ESCALA

REVISIÓN

	NOMBRE	FIRMA	FECHA
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N.º DE DIBUJO

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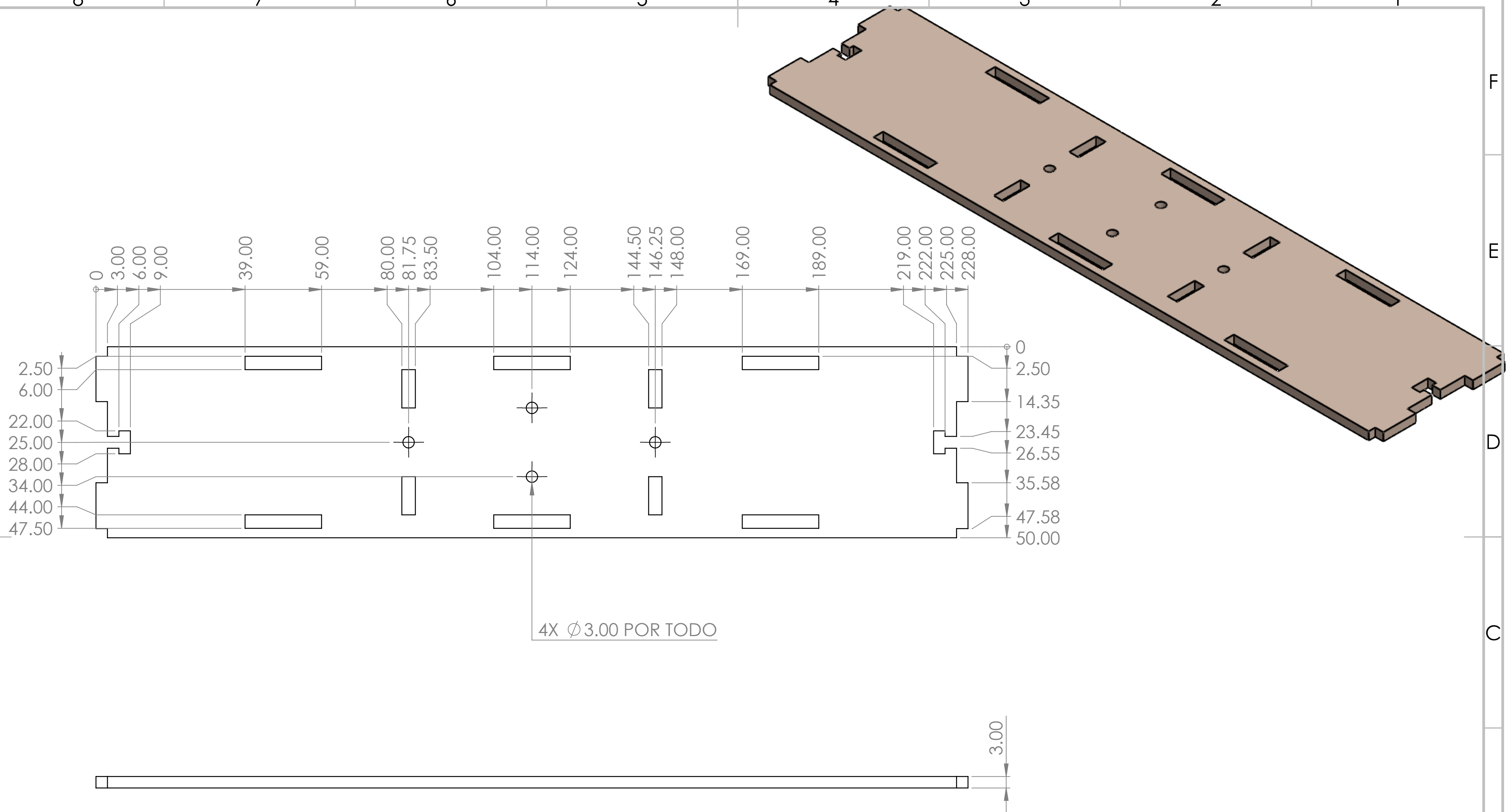
ESCALA:1:2

HOJA 1 DE 1



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	NOMBRE	FIRMA	FECHA			TÍTULO:
DIBUJ.						
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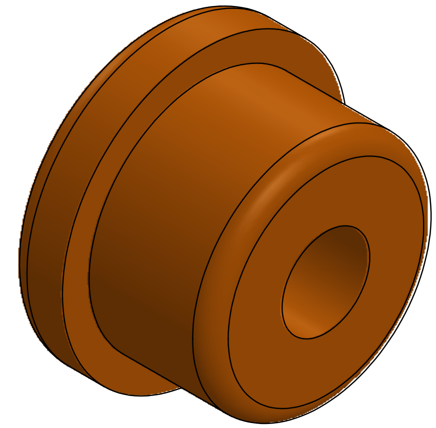

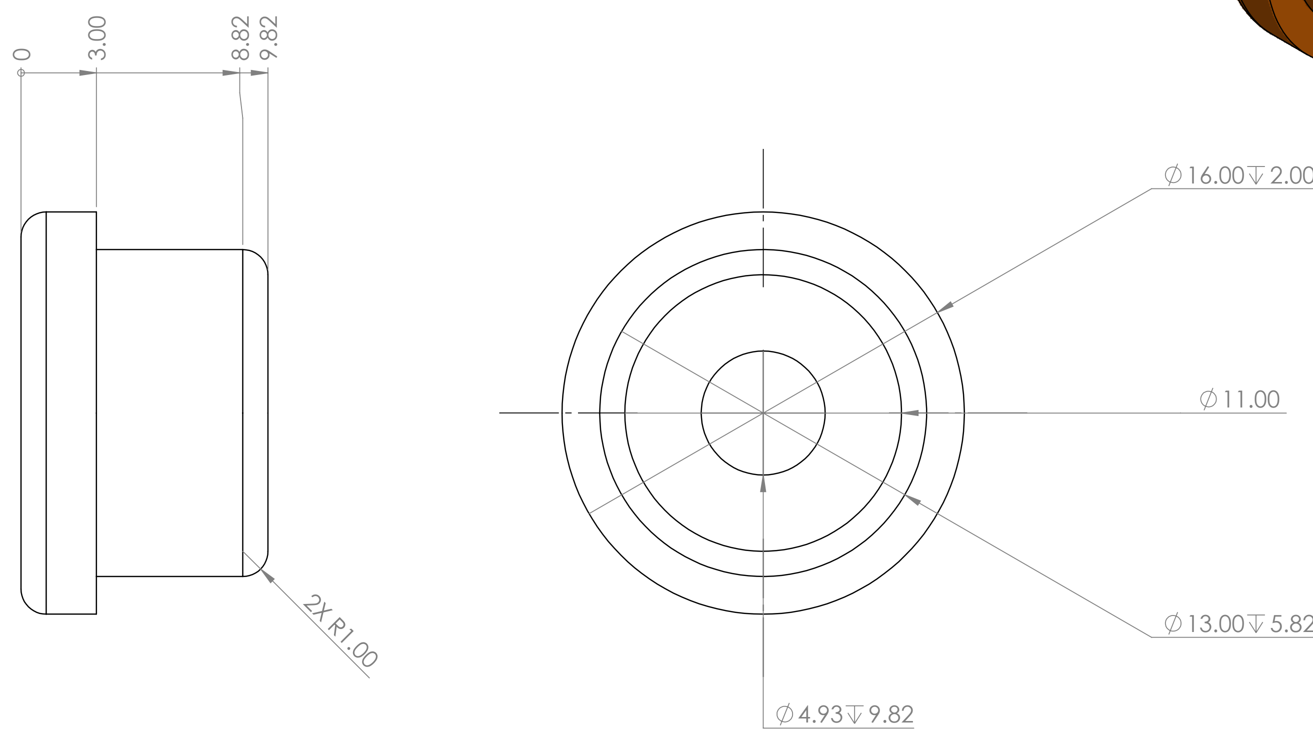

SI NO SE INDICA LO CONTRARIO:
LAS COTAS SE EXPRESAN EN MM
ACABADO SUPERFICIAL:
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LINEAL:
ANGULAR:

ACABADO:

REBARBAR Y ROMPER ARISTAS VIVAS

NO CAMBIE LA ESCALA

REVISIÓN

	NOMBRE	FIRMA	FECHA
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VERIF.			
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CALID.			

TÍTULO:

MATERIAL:

N.º DE DIBUJO

Bujepr_CNC

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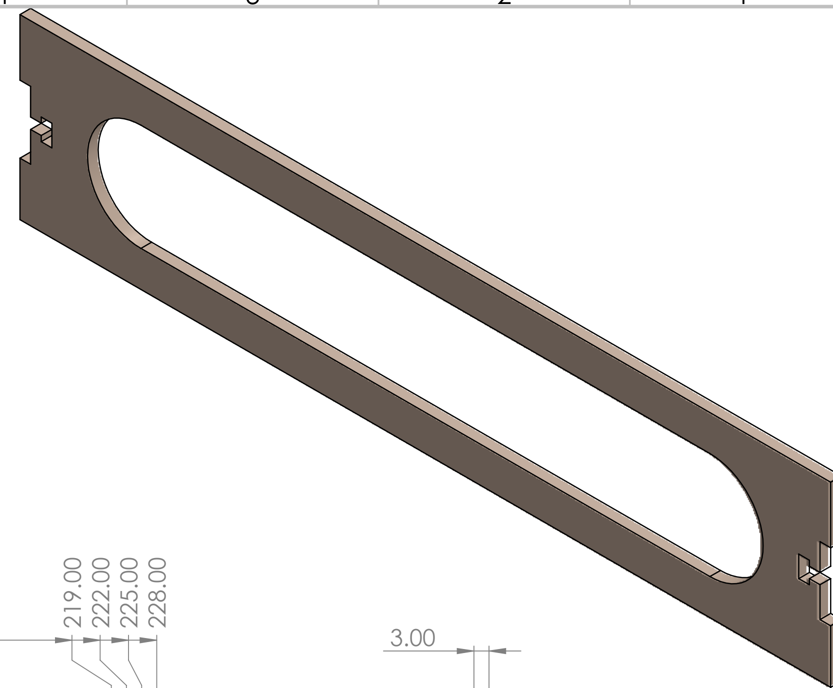
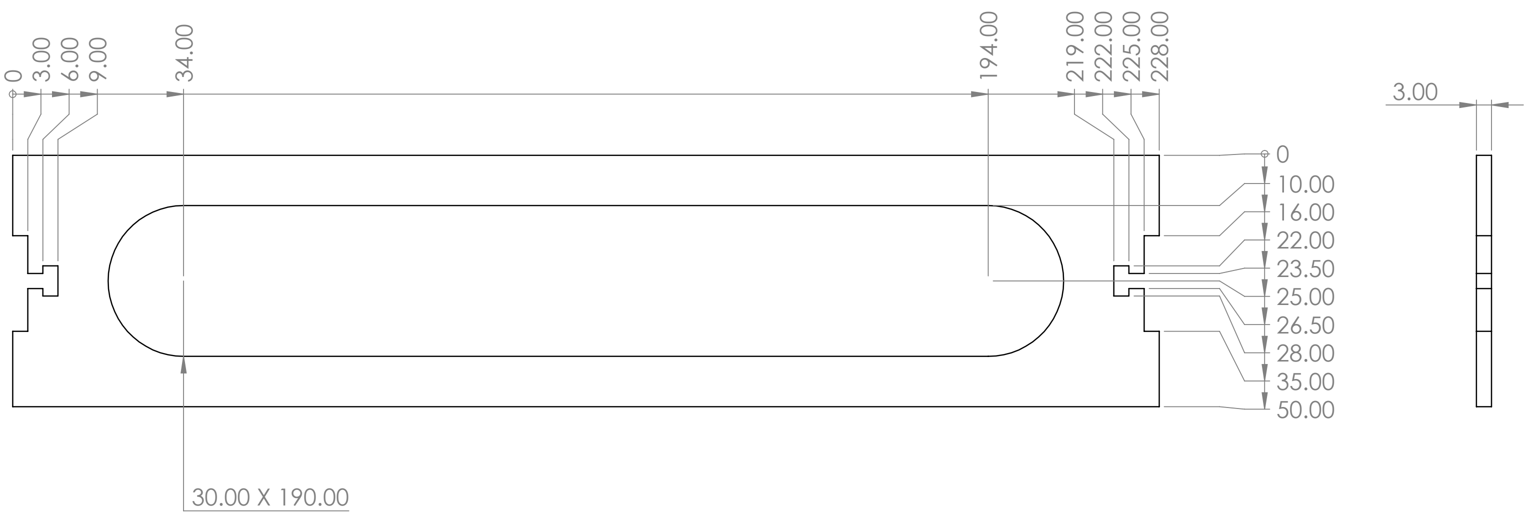
PESO:

ESCALA:5:1

HOJA 1 DE 1





SI NO SE INDICA LO CONTRARIO:
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REBARBAR Y ROMPER ARISTAS VIVAS

NO CAMBIE LA ESCALA

REVISIÓN

	NOMBRE	FIRMA	FECHA
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VERIF.			
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CALID.			

MATERIAL:

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TÍTULO:

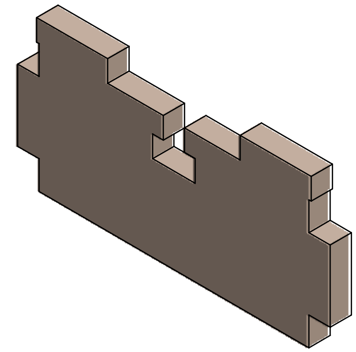
N.º DE DIBUJO

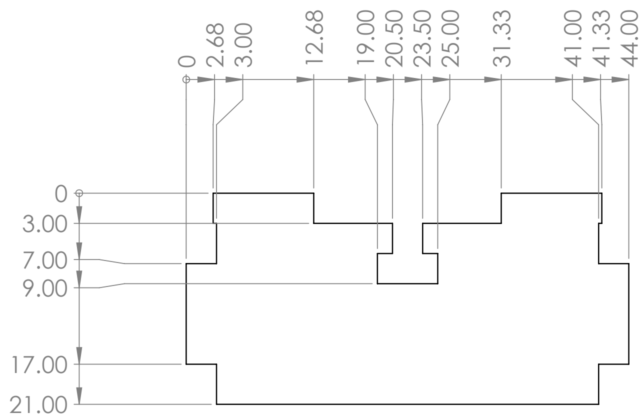
FronteY_CNC

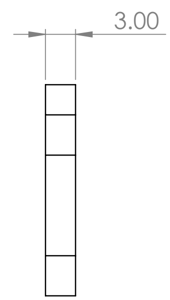
A3

ESCALA:1:2

HOJA 1 DE 1







SI NO SE INDICA LO CONTRARIO:
LAS COTAS SE EXPRESAN EN MM
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TOLERANCIAS:
LINEAL:
ANGULAR:

ACABADO:

REBARBAR Y ROMPER ARISTAS VIVAS

NO CAMBIE LA ESCALA

REVISIÓN

	NOMBRE	FIRMA	FECHA
DIBUJ.			
VERIF.			
APROB.			
FABR.			
CALID.			

TÍTULO:

MATERIAL:

N.º DE DIBUJO

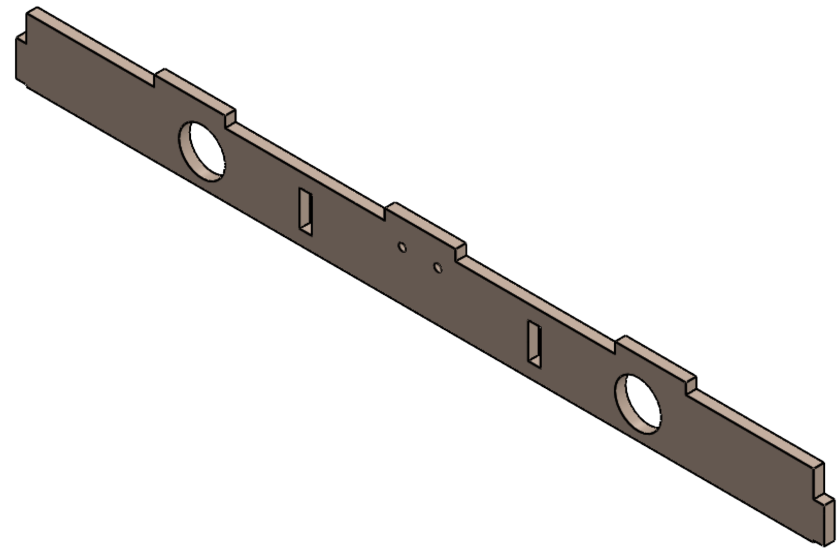
SoportePequeno_CNC

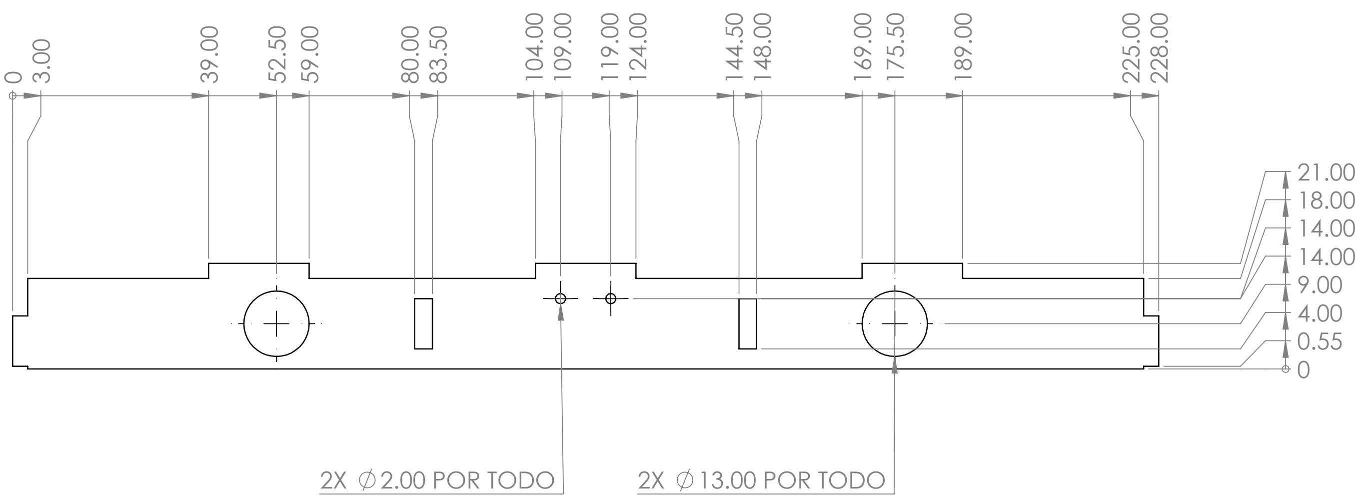
A3

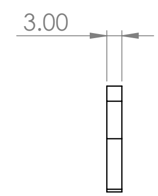
PESO:

ESCALA:2:1

HOJA 1 DE 1







SI NO SE INDICA LO CONTRARIO: LAS COTAS SE EXPRESAN EN MM ACABADO SUPERFICIAL: TOLERANCIAS: LINEAL: ANGULAR:			ACABADO:	REBARBAR Y ROMPER ARISTAS VIVAS	NO CAMBIE LA ESCALA	REVISIÓN
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DIBUJ.						
VERIF.						
APROB.						
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CALID.				MATERIAL:	N.º DE DIBUJO SoporteBuje_CNC	A3
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